



# Gait recognition based on attention convolutional networks and part-based feature representation

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## ABSTRACT

Gait recognition systems are often affected by factors such as changes in viewing angle, clothing, carried items, and complex backgrounds, leading to significant intra-class variations in gait features and consequently reducing recognition performance. Therefore, how to construct a gait feature representation with strong discriminative power and good robustness has become a key issue in current gait recognition research. To address these issues, this paper proposes a gait recognition method that combines an attention-based convolutional network with block-based feature modeling. First, a gait representation is constructed using gait contour sequences, and global spatial features are extracted using a convolutional neural network. Concurrently, a Convolutional Block Attention Module (CBAM) is introduced to enhance the model's ability to focus on key human motion regions (such as the legs and joints), thereby improving feature expressiveness. Subsequently, the gait feature map is structurally partitioned using a region-based chunking strategy to extract finer-grained local features. A multi-channel feature fusion strategy is then employed to integrate global and local information, thereby constructing a more robust gait feature representation. Finally, a classification network is used to perform identity recognition on the fused features, achieving high-precision gait recognition. We conducted extensive experiments on the CASIA-B gait dataset and systematically evaluated the model's performance across different viewpoints and various walking conditions (normal walking, carrying objects, and wearing a jacket). The experimental results demonstrate that the proposed method outperforms multiple baseline models in metrics such as Rank-1 accuracy and mAP, and exhibits superior stability and generalization capabilities across different viewpoints.

## 1. Introduction

With the rapid development of computer vision and artificial intelligence technologies, biometric recognition technologies have found widespread application in fields such as intelligent surveillance, public safety, and smart cities. Among the various biometric technologies, gait recognition has gradually emerged as an important means of identity verification due to its ability to operate at a distance, its contactless nature, and the fact that it requires no user cooperation [1]. Compared to traditional biometric technologies such as fingerprint and facial recognition, gait recognition enables the identification of individuals in remote surveillance environments, making it highly valuable in scenarios such as public safety monitoring, intelligent transportation, and behavioral analysis. For example, in video surveillance systems, even when a subject's facial features cannot be clearly captured, identity verification can still be achieved by analyzing their gait patterns, which broadens the application prospects of gait recognition in complex environments [2].

The core concept of gait recognition is to distinguish between individuals by utilizing the unique movement patterns and structural characteristics formed during human walking. Because the human skeletal structure, body proportions, and movement habits exhibit a certain degree of stability, different individuals display unique dynamic patterns when walking [3,4]. Therefore, by analyzing gait sequences, it is possible to construct feature representations capable of discrimination. However, gait recognition tasks in real-world scenarios still face numerous challenges. For example, factors such as changes in camera angle, clothing, carried items, and environmental noise can significantly affect gait features, thereby reducing recognition accuracy. Furthermore, gait images captured from different camera angles exhibit substantial differences in their structural appearance, making cross-view gait recognition a critical research issue. Consequently, extracting gait features that are both stable and discriminative, while improving the model's generalization ability in complex environments, has become a key focus of current gait recognition research [5].

In recent years, with the advancement of deep learning technologies,

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methods based on deep neural networks have made significant progress in gait recognition tasks. Traditional gait recognition methods typically rely on manually designed features, such as gait energy images (GEI), contour features, and human keypoints. These methods construct gait feature representations by performing statistical analysis on sequences of human contours [6,7]. However, manually designed features have limited expressive power in complex environments and struggle to adapt to variations in multiple covariates. Consequently, an increasing number of studies have begun to utilize deep learning methods to automatically learn gait features. For example, Convolutional Neural Networks (CNNs) can extract multiscale spatial features through their hierarchical structure, thereby significantly improving gait recognition performance. Additionally, some studies have introduced attention mechanisms to enable models to focus more on key areas of human motion, thereby enhancing feature expressiveness [8]. Gollavilli et al. [9] propose a cloud-based hybrid Gradient Boosting–CNN framework for cardiovascular disease prediction using scalable feature learning. Inspired by this multi-model fusion and deep feature extraction approach, the proposed method incorporates hybrid learning and feature integration strategies.

In existing research, various types of gait representation methods have been widely adopted. For example, contour-based gait representation describes structural features of gait by extracting binary human body contours; skeleton-based gait representation, on the other hand, utilizes human keypoints and their connectivity to characterize human motion patterns; additionally, some studies employ 3D human body modeling to obtain more comprehensive gait information [10,11]. These methods have achieved certain results in different scenarios. However, due to differences in how information is represented, the applicability of these methods varies. For instance, contour features perform well in environments with simple backgrounds but are susceptible to noise in complex scenes; while skeleton features better describe the structure of human movement, errors in keypoint detection may degrade recognition performance. Consequently, how to fuse multiple feature types to construct a more robust gait representation has become a key focus of current research [12,13]. The contour-based representation for gait analysis has been used in this project because of its efficient computation and its ability to model the overall body shape and periodic movement in a fairly simple manner. However, there exist several other representations of gait such as the skeleton-based representation and 3D representation. The main advantage of the skeleton-based approach is that it explicitly models the joint dynamics and is therefore able to represent fine grained movements well; however, its efficacy strongly depends upon accurate detection of keypoints, which is very difficult under conditions of occlusions or variation in viewpoints. The 3D representation, on the other hand, is invariant to different viewpoints and provides richer spatial information about motion, but is very complex and computationally expensive to generate.

On the other hand, with the continuous development of deep neural network architectures, some studies have begun to explore multi-model fusion and feature fusion strategies. For example, by combining different deep neural network architectures (such as ResNet, Vision Transformer, or EfficientNet), richer feature representations can be obtained. In addition, some studies utilize attention mechanisms to enhance information in key regions and achieve multi-scale feature learning through multi-channel network architectures [14]. Recent studies in neurological disorder diagnosis have used rule-based systems, machine learning, and deep learning for early detection and improved classification. However, existing methods still have limitations. Many approaches rely on single-modality or rule-based systems, limiting their ability to capture complex neurological symptoms. In addition, most methods do not effectively integrate multimodal data such as imaging and sensor signals, leading to weaker feature representation. Furthermore, variations in patient data across sources reduce model generalization in real-world clinical settings [15].

These methods have improved gait recognition performance to some extent. However, existing research still has some shortcomings. First,

many methods rely primarily on a single feature representation, making it difficult to fully leverage the complementary relationship between human appearance and motion information. Second, during the feature fusion process, some methods fail to effectively eliminate redundant features, thereby increasing model complexity and affecting recognition efficiency. Furthermore, in complex environments and cross-dataset scenarios, gait feature distributions vary significantly, and the model's generalization ability still needs improvement [16]. Gait recognition systems using classical techniques like Gait Energy Image (GEI) describe gait through the process of averaging silhouette information across an entire walking cycle. While being computationally efficient, GEI has limited capability in distinguishing gait characteristics based on spatial and motion details, making it highly susceptible to changes in viewpoints, clothing, and occlusions. In order to overcome such challenges, CBAM is employed in this paper to enhance gait recognition through deep learning. This method allows for adaptive attention to discriminable parts of gait features, specifically limbs and motion regions.

To address the aforementioned issues, this paper proposes a gait recognition method based on dual-branch transfer learning and feature fusion optimization. By combining appearance information with motion information, this method achieves a more comprehensive modeling of gait features. Specifically, this paper first uses an optical flow algorithm to extract human motion information and obtains a more stable representation of the motion region through region cropping. Subsequently, a dual-branch feature extraction network based on transfer learning is constructed, with the two branches dedicated to extracting appearance features and motion features, respectively. By incorporating a pre-trained MobileNetV2 network and fine-tuning it, the model can rapidly learn discriminative gait features even with a limited dataset. After obtaining the dual-branch features, this paper further employs a high-dimensional feature fusion strategy to integrate different feature representations, thereby enhancing the model's expressive power. To reduce redundant features and improve classification performance, this paper introduces an improved Monarch Butterfly Optimization (MBO) algorithm for feature selection, obtaining an optimal feature subset through an optimization search to enhance the model's recognition accuracy.

Existing gait recognition methods can be broadly grouped into three categories based on feature representation: contour-based, skeleton-based, and optical flow-based approaches. Contour-based methods, such as GEI-based representations but they are sensitive to occlusion and clothing variations. Skeleton-based methods utilize human joint information and structural pose modeling. However, their performance depends heavily on accurate pose estimation. Optical flow-based methods capture motion dynamics between consecutive frames, but they are computationally expensive and sensitive to background noise. Despite these advancements, cross-view gait recognition remains a major challenge, as significant performance degradation occurs when training and testing viewpoints differ due to large appearance and motion variations.

## 2. Methods

### 2.1. Overall framework

The overall framework of the gait recognition method proposed in this paper is shown in Fig. 1.

The input ff-GEI representation captures the spatial distribution of gait over time. The model processes these inputs through a multi-channel convolutional network enhanced with CBAM attention to focus on key motion regions. Additionally, a regional partition strategy is applied to extract fine-grained local features, which are later fused with global features for robust gait representation. The model consists of three main components: 1) a multi-channel convolutional block attention feature extraction network; 2) a regional feature extraction module; 3) a gait classification module. The model first takes frame-wise Gait Energy Images (ff-GEI) as input and extracts gait features through a

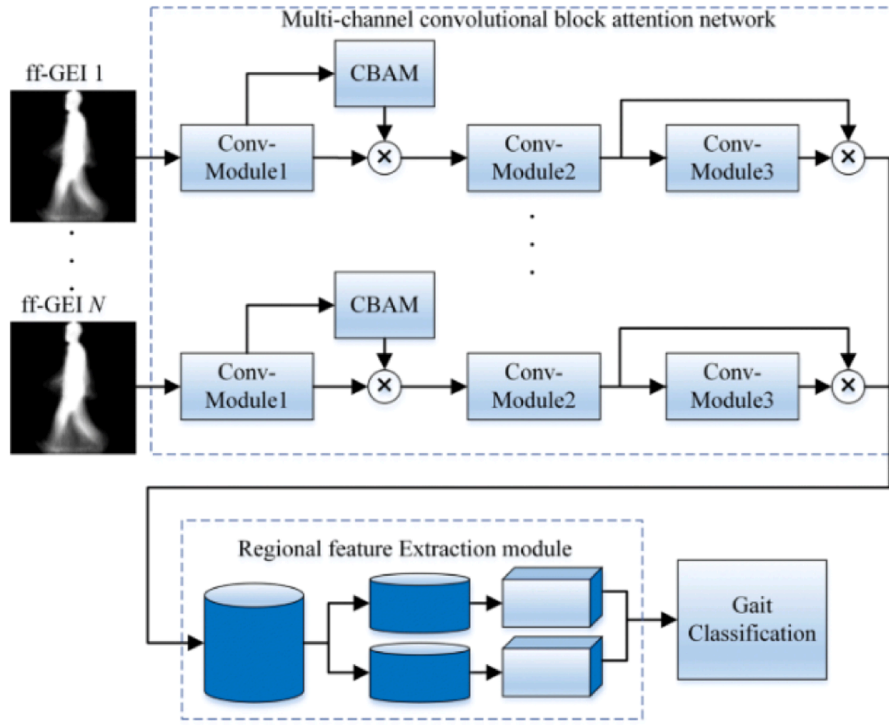


Fig. 1. The proposed scheme in this paper.

multi-channel convolutional network. The CBAM attention mechanism is introduced during the convolutional feature extraction process, enabling the model to adaptively focus on key motion regions within the gait images, thereby enhancing feature representational capacity. The multi-channel convolutional modules first extract feature maps from the input gait representations. These feature maps are then passed to the CBAM module, which sequentially applies channel attention followed by spatial attention. The channel attention mechanism learns to emphasize informative feature channels by modeling inter-channel dependencies, while the spatial attention mechanism focuses on the most discriminative spatial regions of the gait silhouette. The output of CBAM is then re-weighted feature maps that are fed into subsequent layers for further processing. This sequential interaction ensures that the convolutional features are adaptively refined in both channel and spatial dimensions, improving the discriminative power of the learned gait representation.

The attention mechanism employed in this study is formally defined based on the Convolutional Block Attention Module (CBAM), which sequentially applies channel and spatial attention to refine feature representations. Given an intermediate feature map  $F \in \mathbb{R}^{C \times H \times W}$ , the channel attention map  $M_c(F)$  is computed as:

$$M_c(F) = \sigma(\text{MLP}(\text{AvgPool}(F)) + \text{MLP}(\text{MaxPool}(F))) \quad (1)$$

where  $\sigma$  denotes the sigmoid activation function, and MLP represents a shared multi-layer perceptron. The feature map is then refined through channel-wise weighting:

$$F' = M_c(F) \otimes F \quad (2)$$

Subsequently, spatial attention is applied to capture informative regions. The spatial attention map  $M_s(F')$  is defined as:

$$M_s(F') = \sigma(f^{7 \times 7}([\text{AvgPool}(F'); \text{MaxPool}(F')])) \quad (3)$$

where  $f^{7 \times 7}$  denotes a convolution operation with a kernel size of  $7 \times 7$ . The final refined feature representation is obtained as:

$$F'' = M_s(F') \otimes F' \quad (4)$$

Through this formulation, the attention mechanism adaptively emphasizes discriminative gait regions while suppressing irrelevant background information, thereby improving the overall feature representation capability of the model.

Subsequently, the regional feature extraction module spatially partitions the feature maps to extract local feature representations of different body parts. Finally, global and regional features are fused and fed into a classifier for identity recognition.

In the upper half of Fig. 1, a multi-channel convolutional block attention network is used to extract deep gait features. The input ff-GEI sequence is fed into multiple parallel channels, each consisting of several convolutional modules (Conv-Module), followed by a CBAM attention module.

The CBAM module enhances feature representations by sequentially applying channel attention followed by spatial attention.

- 1) The input feature map is first processed using global average pooling and global max pooling to generate two channel descriptors. These descriptors are passed through a shared multilayer perceptron (MLP) to learn channel-wise importance. The resulting attention weights are applied to the input feature map to emphasize informative channels while suppressing less useful ones.
- 2) The refined feature map from the channel attention stage is then forwarded to the spatial attention module. Average pooling and max pooling are applied along the channel dimension, and the resulting feature maps are concatenated. A convolution operation is then used to generate a spatial attention map, which highlights important spatial regions.

The convolutional modules are primarily used to learn spatial structural features in gait images, while the attention modules enhance the response in key regions by weighting the channel and spatial dimensions. In gait recognition tasks, the human leg movement region typically contains richer dynamic information; therefore, the attention mechanism can automatically emphasize the importance of these regions while suppressing background noise or irrelevant information. Through the multi-channel structure, different channels can learn gait

features at different scales or with different patterns. The features extracted by each channel are fused in the back end of the network, thereby forming a richer gait representation. Different deep learning models offer diverse features due to their unique learning principles. For example, VGG-16 is mainly concerned with the lower-level spatial patterns and edges. On the other hand, ResNet-50 learns the higher-level spatial structures. The Vision Transformers model captures the relationships across gait sequences. Meanwhile, EfficientNet-B0 is capable of capturing features at different scales efficiently. By combining these diverse features, the proposed network able to gain multiple views of the gait information. This results in the creation of more robust features while minimizing bias towards any specific feature type.

To further enhance the model's ability to capture fine-grained gait information, this paper introduces a regional feature extraction module following the convolutional feature extraction stage. As shown in the lower half of the figure, this module first receives high-level feature maps from the convolutional network and then divides the feature maps along the vertical axis of the human body to obtain multiple regional features. This method of regional division enables the capture of distinct motion patterns in different parts of the human body [17]. For example, the upper-body region primarily contains relatively stable contour information, while the lower-body region contains richer dynamic gait features. By extracting these regional features separately and combining them, the model can obtain a more discriminative representation of gait.

Let the feature map output by the convolutional network be denoted by (F). The process of local feature extraction can be expressed as:

$$F_{part} = f_1, f_2, \dots, f_K \quad (5)$$

Where, K denotes the number of partitioned regions, and  $f_K$  denotes the feature vector of the k-th region. By applying pooling operations to each region, the corresponding local feature representations can be obtained.

After obtaining the regional features, the model fuses them with the global features extracted by the convolutional network to produce the final gait representation vector. The fused features contain not only overall gait information but also local motion patterns, thereby better capturing an individual's walking characteristics. Finally, the fused features are fed into the gait classification module, where the identity recognition task is performed using a fully connected layer and the Softmax function.

To train the proposed gait recognition model, this paper adopts the cross-entropy loss function as the optimization objective. Let the training dataset be  $(x_i, y_i)_{i=1}^N$ , where  $x_i$  represents an input gait sample and  $y_i$  represents the corresponding identity label. The category probability predicted by the model is  $p(y_i|x_i)$ .

The overall optimization objective of the model can be expressed as:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log p(y_i|x_i) \quad (6)$$

Where, N denotes the number of training samples, and  $p(y_i|x_i)$  denotes the model's predicted probability that sample  $x_i$  belongs to class  $y_i$ . By minimizing this loss function, the model can learn more discriminative representations of gait features, thereby improving the accuracy of gait recognition.

The proposed architecture is composed of three distinct components: (1) an attention-based convolutional network enhanced with CBAM for spatial and channel-wise feature refinement, (2) a multi-network feature extraction module that integrates complementary representations from multiple deep learning architectures, and (3) a dual-branch MobileNetV2 framework designed to separately capture appearance and motion features. These components operate at different stages of the pipeline and serve complementary roles, thereby avoiding conceptual overlap while enhancing the overall feature representation capability.

## 2.2. Specific implementation of the model

The multi-network fusion module, this component is specifically designed to model appearance and motion information using a two-branch architecture based on MobileNetV2. As shown in Fig. 2, the gait recognition model proposed in this paper effectively models gait information through multi-network feature extraction and a multi-level fusion strategy. The overall architecture consists of three main components: a multi-model feature extraction network, a multi-level fusion module, and a final classification decision module [18]. This framework enables parallel learning of gait features by incorporating various deep learning network architectures, thereby fully leveraging the strengths of different networks in feature representation. Additionally, it enhances recognition stability and accuracy through feature-level and decision-level fusion strategies. Feature-level fusion refers to combining representations extracted from different branches before classification, enabling joint learning of complementary spatial and motion information. In contrast, decision-level fusion combines the final classification outputs from multiple models or branches, aggregating decisions to improve robustness. The proposed framework primarily adopts feature-level fusion to preserve rich representational information before classification, ensuring more discriminative gait features.

The multi-network fusion stage is very thoughtful and aims at making sure that diverse and complementary feature representations can be captured. The use of VGG-16 stems from the simplicity and consistency of the model when it comes to convolution. On the other hand, ResNet-50 has been chosen because of its ability to learn features via residual learning techniques. In addition, GoogLeNet incorporates multiple scales of feature extraction thanks to its inception modules, while the EfficientNet-B0 strikes a balance between accuracy and computation. Moreover, Vision Transformer is used to take care of the global relationship of gait features via self-attention. The use of such a heterogeneous network architecture guarantees that the feature representation.

Deep learning models are employed to capture complementary gait features at multiple scales and abstraction levels. The extracted features are integrated using a feature-level fusion mechanism, followed by decision-level aggregation to enhance recognition robustness. First, in the feature extraction stage, this paper employs five typical deep neural networks as feature extractors, including VGG-16, Vision Transformer, ResNet-50, GoogLeNet, and EfficientNet-B0. The input data consists of frame-by-frame gait energy image (ff-GEI) sequences. For a given gait sample (x), each network extracts corresponding deep feature representations. Since these networks differ in terms of receptive fields, feature abstraction capabilities, and spatial modeling approaches, they are able to capture discriminative information in gait images from different perspectives. Let there be a total of (M) feature extraction networks; then, the features output by the (i)-th network can be expressed as

$$F_i = f_i(x; \theta_i), \quad i = 1, 2, \dots, M \quad (7)$$

Where,  $f_i(\cdot)$  denotes the feature mapping function of the ith deep neural network,  $\theta_i$  represents the corresponding network parameters, and  $F_i$  denotes the extracted gait feature representation. Through this parallel feature learning approach, richer and more complementary gait information can be obtained.

After obtaining multi-model feature representations, this paper designs a multi-level fusion mechanism to integrate feature information from different networks. First, in the feature-level fusion stage, the feature vectors extracted by multiple networks are weighted and fused to obtain a unified feature representation. This fusion process can be expressed as

$$F_{\text{fusion}} = \sum_{i=1}^M \alpha_i F_i \quad (8)$$

Where,  $F_i$  represents the output features of the ith network,  $\alpha_i$  denotes



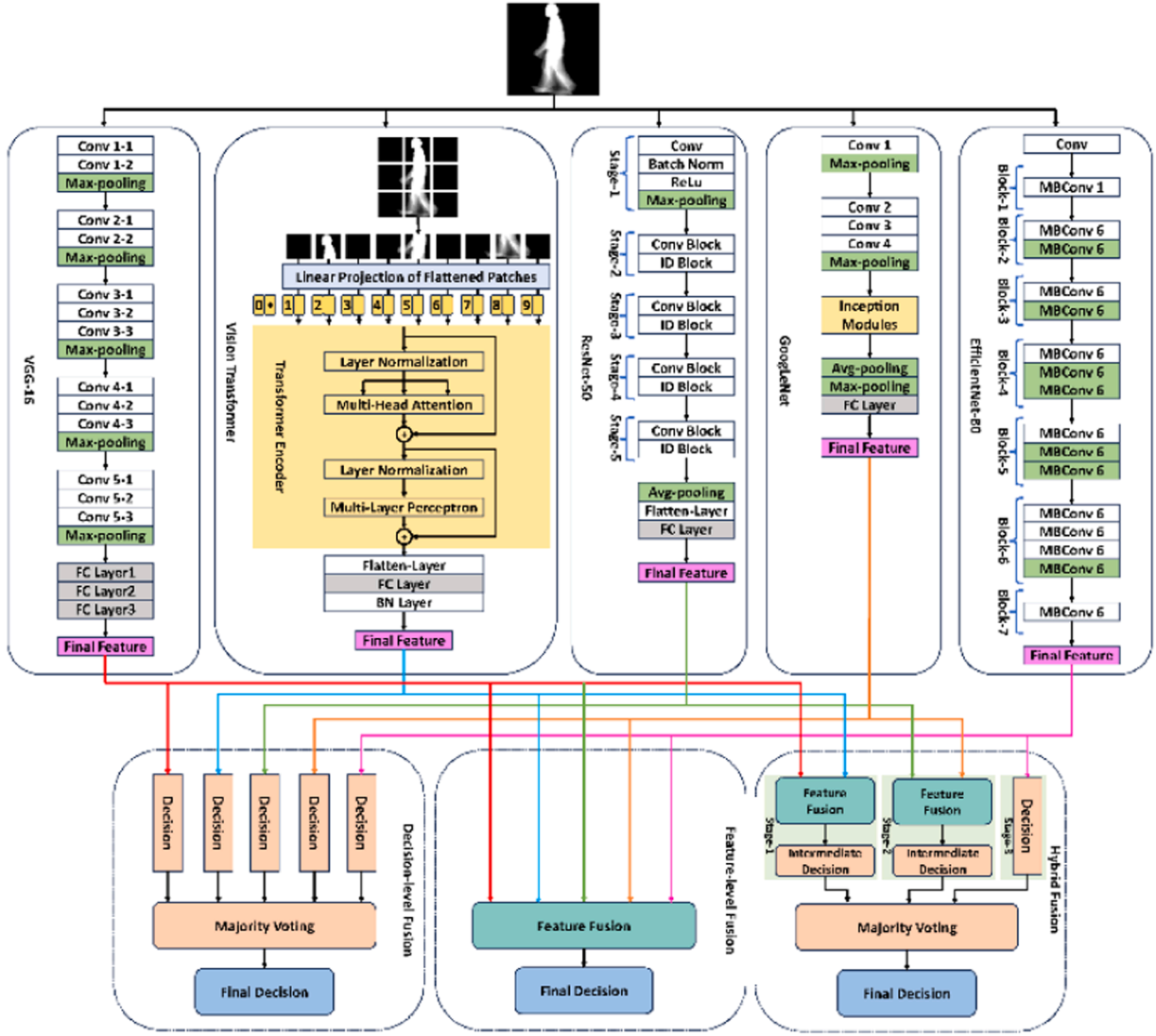


Fig. 2. Multi-network fusion gait recognition.

the corresponding feature weight coefficients, and  $F_{fusion}$  denotes the fused gait feature vector. This approach allows for the combined utilization of the strengths of different networks, thereby enhancing the discriminative power of the feature representation.

In addition to feature-level fusion, this paper also proposes a decision-level fusion strategy. At this stage, each feature extraction network independently performs the classification task and outputs its corresponding prediction results. Subsequently, the multiple classification results are fused using a majority voting mechanism to obtain the final recognition result [19,20]. This strategy mitigates the impact of prediction errors from individual models to some extent, thereby improving the overall system stability. Furthermore, to further enhance model performance, this paper proposes a hybrid fusion strategy that combines feature-level fusion with decision-level fusion, enabling the model to simultaneously utilize information from both the feature layer and the classification layer.

During training, the fused feature representations are fed into the classifier for classification. The model uses the Softmax function to output the probability distribution of a sample belonging to each category and is optimized using the cross-entropy loss function. Let the training dataset be  $(x_i, y_i)_{i=1}^N$ , where  $x_i$  represents the input gait sample and  $y_i$  is the corresponding identity label. Then, the overall optimization objective of the model can be expressed as

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log P(y_i | x_i) \quad (9)$$

Where,  $P(y_i | x_i)$  represents the model's predicted probability that sample  $x_i$  belongs to class  $y_i$ . By minimizing this loss function, the model can learn gait feature representations with strong discriminative power, thereby achieving accurate identity recognition.

In summary, the model proposed in this paper fully utilizes gait information through parallel feature extraction across multiple networks, multi-level feature fusion, and a decision-level voting mechanism. This method not only enhances the expressive power of gait features but also effectively improves the robustness of the recognition system in complex environments.

### 2.3. Feature extraction and fusion in multi-branch transfer learning

This module operates independently of the dual-branch structure and focuses on extracting complementary features from multiple deep models.

Fig. 3 shows the dual-branch transfer learning pipeline used in the proposed method. One branch processes the appearance information derived from gait energy images, while the other branch captures motion information using optical flow-based representations. Both

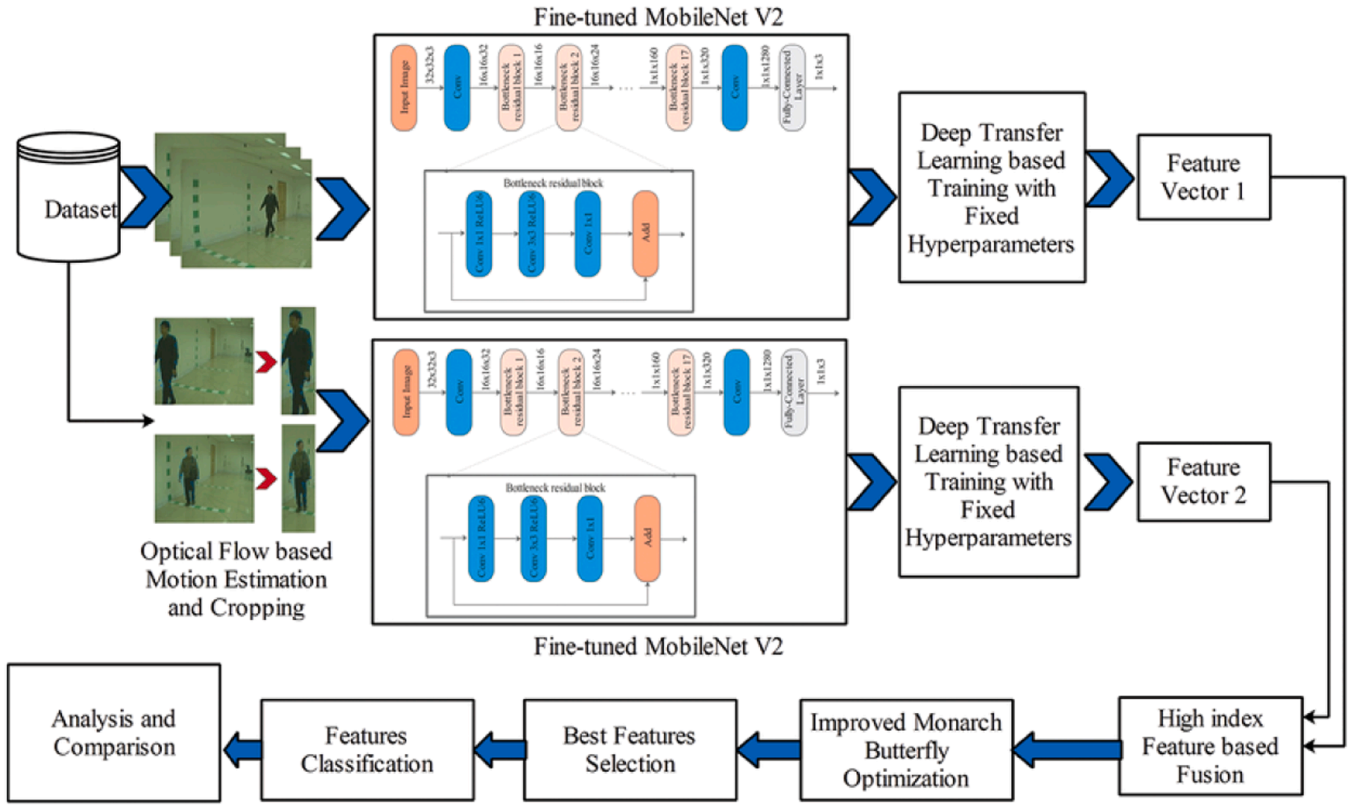


Fig. 3. Gait recognition using a convolutional attention network combined with block-based features.

branches utilize a fine-tuned MobileNetV2 network to extract high-level features, which are subsequently fused to form a comprehensive gait representation. The overall workflow includes: data input and pre-processing, motion estimation and cropping based on optical flow, dual-branch MobileNetV2 feature extraction, feature fusion, optimal feature subset selection based on MBO, and final classification and performance analysis. This architecture simultaneously utilizes appearance and motion information to obtain a more discriminative gait representation.

Let the input gait video sequence be

$$\mathcal{V} = I_1, I_2, \dots, I_T \quad (10)$$

Where,  $I_t \in \mathbb{R}^{H \times W}$  denotes the  $t$ -th frame of the image, and  $T$  is the sequence length. To reduce the dimensionality of the data and highlight human structural information, we first construct a gait energy representation. For a given sequence, its gait representation can be defined as

$$G(x, y) = \frac{1}{T} \sum_{t=1}^T I_t(x, y) \quad (11)$$

Where,  $G(x, y)$  represents the gait energy index (GEI), which describes the average distribution of the human body contour over a single gait cycle.

To enhance the modeling of motion information, this paper uses an optical flow algorithm to calculate the motion changes between adjacent frames. Let the images of frames  $t$  and  $t + 1$  be denoted as  $I_t$  and  $I_{t+1}$ , respectively; the optical flow vector field can be expressed as

$$\mathbf{u}(x, y) = (u(x, y), v(x, y)) \quad (12)$$

Where,  $u(x, y)$  and  $v(x, y)$  represent the horizontal and vertical components of motion, respectively. Based on the assumption of constant brightness, we obtain the classical optical flow constraint equation

$$I_x u + I_y v + I_t = 0 \quad (13)$$

Where  $I_x$  and  $I_y$  denote spatial gradients along horizontal and vertical directions,  $I_t$  represents temporal intensity variation, and the equation

enforces motion consistency between consecutive frames.

The regions of human motion are calculated based on the optical flow magnitude:

$$M(x, y) = \sqrt{u(x, y)^2 + v(x, y)^2} \quad (14)$$

Extract and crop the region of motion based on the threshold:

$$R = (x, y) | M(x, y) > \tau \quad (15)$$

This step reduces background noise and highlights areas of human motion.

To simultaneously utilize both appearance and motion features, this paper constructs a dual-branch feature extraction network. Both branches use a fine-tuned MobileNetV2 as their base network; one branch takes the original gait image as input, while the other takes the motion image cropped using optical flow [21,22].

Let the input sample be  $(x)$ ; the features extracted by the  $(i)$ th network are represented as

$$F_i = f_i(x; \theta_i) \quad (16)$$

Where,  $f_i(\cdot)$  denotes the  $i$ th MobileNetV2 network, and  $\theta_i$  represents its parameters. This paper consists of two branches:

$$F_1 = f_1(x_{\text{appearance}}) \quad (17)$$

$$F_2 = f_2(x_{\text{motion}}) \quad (18)$$

Where,  $F_1$  denotes the appearance feature vector (Feature Vector 1), and  $F$  denotes the motion feature vector (Feature Vector 2).

To effectively utilize both types of feature information, this paper employs a high-dimensional feature fusion strategy. Let the two sets of features be denoted as  $F_1 \in \mathbb{R}^{(d_1)}$  and  $F_2 \in \mathbb{R}^{(d_2)}$ , respectively. Then, the fused feature is expressed as

$$F_{\text{fusion}} = \alpha F_1 \oplus \beta F_2 \quad (19)$$

Where  $F_{fusion}$  denotes the final fused feature representation,  $F_1$  represent features from two different branches (e.g., appearance and motion),  $\alpha$  and  $\beta$  are weighting coefficients controlling their contributions, and  $\oplus$  indicates the fusion operation (such as concatenation or element-wise combination). The dimension of the fused features is

$$d = d_1 + d_2 \quad (20)$$

This high-dimensional feature vector simultaneously incorporates both human appearance and motion information, thereby enhancing the expressive power of the features.

The interpretability and provide a clear formulation of the feature integration process, a weighted feature fusion strategy is adopted to combine global and local gait representations. Let  $F_g$  and  $F_l$  denote the global and local feature vectors, respectively. The final fused representation  $F_f$  is defined as:

$$F_f = \alpha F_g + (1 - \alpha) F_l \quad (21)$$

where  $\alpha \in [0,1]$  is a learnable or empirically tuned weighting coefficient that controls the contribution of each feature type. This formulation ensures a balanced integration of complementary information, where global features capture holistic gait structure and local features preserve fine-grained motion details. By explicitly modeling the contribution weights, the fusion mechanism enhances interpretability and supports reproducibility across different experimental settings.

To reduce feature redundancy and improve classification performance, this paper employs an improved Monarch Butterfly Optimization algorithm for feature selection. The Monarch Butterfly Optimization (MBO) algorithm is inspired by the migration behavior of monarch butterflies and balances exploration and exploitation through migration and butterfly adjusting operators. The migration operator enables global exploration by allowing candidate solutions to move across regions, helping to avoid premature convergence, while the butterfly adjusting operator enhances local exploitation through solution refinement. Compared to Genetic Algorithms, which rely on crossover and mutation, MBO employs a structured migration mechanism that improves diversity in global search. Similarly, unlike Particle Swarm Optimization, which uses velocity-based updates, MBO adopts a probabilistic migration strategy that enhances exploration in complex feature spaces.

Let the set of fused features be  $F_{fusion}$ ; the task is to find the optimal subset of features  $S \subseteq F_{fusion}$ . The optimization objective is

$$minSJ(S) = \lambda E(S) + (1 - \lambda) \frac{|S|}{|F|} \quad (22)$$

Where:  $E(S)$  is the classification error when using the feature subset  $(S)$ ,  $(|S|)$  is the number of selected features,  $(|F|)$  is the total number of features, and  $\lambda$  is the balancing parameter

The Monarch Butterfly Optimization algorithm is used to search the feature space to minimize the classification error while keeping the feature dimension as low as possible.

The optimal feature subset selected is then fed into the classifier for identity recognition. The classifier uses a Softmax model, whose output probability is

$$P(y|x) = \frac{e^{W_y F + b_y}}{\sum_{k=1}^C e^{W_k F + b_k}} \quad (23)$$

Where  $P(y|x)$  denotes the probability of class  $y$  given input  $x$ ,  $F$  represents the extracted feature vector,  $C$  is the total number of classes, and the denominator ensures normalization across all classes using the softmax function., and  $W_k$  and  $b_k$  are the classifier parameters.

The model is trained by minimizing the cross-entropy loss, and its overall optimization objective is

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log P(y_i | x_i) \quad (24)$$

where  $(N)$  is the number of training samples and  $y_i$  is the true label.

## 2.4. Model implementation

Integrating the designs of the aforementioned modules, the gait recognition model proposed in this paper constructs a comprehensive recognition framework through dual-branch feature learning, multi-stage feature fusion, and an optimization-driven feature selection mechanism. The overall process is illustrated in the figure and can be summarized into four main stages: data preprocessing and motion information extraction, dual-branch transfer learning-based feature extraction, high-dimensional feature fusion and feature optimization selection, and final classification decision-making [23,24].

First, during the data preprocessing stage, raw video sequences are read from the gait dataset, and standardized gait image sequences are obtained through image preprocessing operations. Concurrently, human motion information is extracted using optical flow motion estimation algorithms to capture dynamic changes between adjacent frames. Motion amplitude analysis is used to identify areas of human motion, followed by region cropping to reduce background interference and highlight human motion features. This process enables the model to simultaneously utilize both static contour information and dynamic motion information, providing a richer data representation for subsequent feature learning.

Subsequently, during the feature extraction stage, this paper adopts a dual-branch network architecture based on transfer learning. Both branches are built on a fine-tuned MobileNetV2 network, with one branch used to extract human appearance contour features and the other to extract optical flow motion features. Through a transfer learning strategy, the network parameters are optimized based on the pre-trained model, enabling the model to adapt more quickly to the data distribution of the gait recognition task. Each branch outputs corresponding high-level feature vectors, thereby forming appearance feature representations and motion feature representations.

After obtaining the dual-branch feature representations, the model proceeds to the feature fusion stage. Through a high-dimensional feature fusion strategy, the feature vectors extracted by the two branches are combined to form a comprehensive feature representation containing multidimensional information. This fused feature not only includes human appearance information but also motion variation information, thereby enabling a more comprehensive description of individual gait characteristics. To further reduce redundant features and enhance feature discriminative power, this paper employs an improved Monarch Butterfly Optimization algorithm to optimize the selection of the fused features. By searching for the optimal feature subset in the feature space, the model obtains a more compact and discriminative feature representation. The suggested gait recognition architecture works on a unified framework where every component contributes effectively towards achieving improved performance through collaboration. The appearance and motion dual-stream feature extraction network captures complimentary features, while the CBAM attention module improves feature discrimination by giving more importance to distinctive gait areas. Multi-channel feature extraction and regional feature extraction contribute towards capturing multi-scale features using enriched spatial-local representations. In the next step, the fusion of the extracted features is achieved in a unified feature representation through the feature fusion layer. Finally, the Monarch Butterfly Optimization technique removes irrelevant features from the unified feature representation through optimization.

Finally, during the classification stage, the optimal feature vector selected through feature selection is fed into the classifier for identity recognition. The classifier outputs the probability of the corresponding class based on the input features and trains the model using the cross-entropy loss function. Through this process, the model gradually learns a gait feature representation with strong discriminative power, thereby achieving accurate identity recognition.

From a holistic perspective, the model proposed in this paper can be represented as a mapping function composed of multiple functional modules. Let the input gait sequence be  $x$ . After undergoing data preprocessing, feature extraction, feature fusion, and feature optimization, the final output is the classification result. This process can be uniformly represented as

$$y = C(O(\phi(f_1(x), f_2(x)))) \quad (25)$$

Where,  $f_1(x)$  and  $f_2(x)$  represent the appearance and motion features extracted by the two-branch MobileNetV2 network, respectively;  $\phi(\cdot)$  denotes the feature fusion function;  $O(\cdot)$  denotes the feature selection process based on Monarch Butterfly Optimization; and  $C(\cdot)$  denotes the final classifier. This unified expression describes the complete computational workflow of the proposed model, from the input gait sequence to the output recognition result.

Overall, by integrating transfer learning, dual-branch feature extraction, multi-level feature fusion, and intelligent optimization algorithms, the proposed model achieves deep modeling of gait information. This framework fully leverages the complementary relationship

between human appearance and motion information while using optimization algorithms to select key features, thereby effectively improving the accuracy and robustness of the gait recognition system. Algorithm 1 illustrates the training and inference workflow of the proposed gait recognition model.

## 2.5. Complexity analysis

To further demonstrate the feasibility of the proposed model in terms of computational and storage costs, this section analyzes the model from the perspectives of time complexity and space complexity. The proposed method primarily consists of four core stages: gait preprocessing and optical flow estimation, dual-branch transfer learning-based feature extraction, high-dimensional feature fusion, and feature selection based on an improved Monarch Butterfly Optimization algorithm. Therefore, the overall complexity is determined by the combined contributions of these components [25].

First, in the preprocessing stage, the input is a gait video sequence of length (T). Assuming a single-frame image size of (H × W), generating

### Algorithm 1

A Gait Recognition Method Based on Bifurcated Transfer Learning and Feature Optimization.

Input:

Gait video sequence dataset  $D = (V_i, y_i)_{i=1}^N$   
 Pre-trained MobileNetV2 parameters  $\theta_{pre}$   
 Light flow threshold  $\tau$   
 Feature fusion weights  $\alpha$ ,  
 Increase the maximum number of iterations  $T_{mbo}$  for the Monarch Butterfly Optimization  
 Initializing classifier parameters  $W, b$

Output:

Trained gait recognition model  $\mathcal{M}$

Optimal feature subset  $S^*$

Predicted label for the sample  $\hat{y}$

- 1: Perform the following on each gait video sequence  $V_i$  in dataset  $D$ :
- 2: Perform frame extraction on  $V_i$  to obtain the image sequence  $I_{t=1}^{T_i}$
- 3: Calculate Gait Energy Index (GEI):  
 $G_i(x, y) = (1/T_i) \sum_{t=1}^{T_i} I_t(x, y)$
- 4: Compute the optical flow field for adjacent frames:  
 $u_t, v_t \leftarrow \text{OpticalFlow}(I_t, I_{t+1})$
- 5: Plot of motion amplitude:  
 $M_t(x, y) = \sqrt{u_t(x, y)^2 + v_t(x, y)^2}$
- 6: Extract the motion region  $R_t$  based on the threshold  $\tau$ , perform cropping, and obtain the motion image  $x_i^{motion}$
- 7: Let GEI or the raw gait representation be denoted as the appearance input  $x_i^{appearance}$
- 8: Initialize a two-branch network:  
 Branch-1  $\leftarrow$  Fine-tuned MobileNetV2( $\theta_{pre}$ )  
 Branch-2  $\leftarrow$  Fine-tuned MobileNetV2( $\theta_{pre}$ )
- 9: Perform two-branch feature extraction on each training sample:
- 10:  $F_i^{appearance} \leftarrow \text{Branch-1}(x_i^{appearance})$
- 11:  $F_i^{motion} \leftarrow \text{Branch-2}(x_i^{motion})$
- 12: Perform high-dimensional feature fusion:  
 $F_i^{fusion} \leftarrow F_i^{appearance} \oplus F_i^{motion}$
- 13: Search for the optimal feature subset in the fused feature space using an improved MBO:
- 14: Initialize the butterfly swarm population  $P = \{S_1, S_2, \dots, S_m\}$
- 15: for  $t = 1$  to  $T_{mbo}$  do
- 16:   Compute the fitness  $J(S_j)$  for each candidate feature subset  $S_j$
- 17:   Update the butterfly swarm's position using migration operators and adjustment operators
- 18:   Retain the current optimal feature subset  $S^*$
- 19: end for
- 20: Select the features corresponding to the optimal feature subset:  
 $F_i^* \leftarrow \text{Select}(F_i^{fusion}, S^*)$
- 21: Feed  $F_i^*$  into the classifier:  
 $p(y_i|x_i) = \text{Softmax}(WF_i^* + b)$
- 22: Calculate the cross-entropy loss:  
 $\theta = -(1/N) \sum_{i=1}^N \log p(y_i|x_i)$
- 23: Update Branch-1, Branch-2, and the classifier parameters using backpropagation
- 24: Repeat steps 9–23 until the convergence criteria are met or the maximum number of training iterations is reached
- 25: Deduction phase:
- 26: Repeat steps 2–12 for the test sample
- 27: Select features using the optimal feature subset  $S^*$
- 28: Input: classifier; Output: predicted label  $y = \arg\max p(y|x)$
- 29: Return the trained model  $\mathcal{M}$ , the optimal feature subset  $S^*$ , and the prediction results  $\hat{y}$



the gait energy map requires summing and averaging each pixel across (T) frames; therefore, its time complexity is

$$\mathcal{O}(THW) \quad (26)$$

In the optical flow estimation process, since motion field calculations must be performed on adjacent frames, if a dense optical flow method is used, the computational complexity for each pair of adjacent frames is typically proportional to the image resolution. Therefore, the computational complexity of optical flow estimation across the entire sequence can be expressed as

$$\mathcal{O}((T-1)HW) \quad (27)$$

Therefore, the overall complexity of the preprocessing stage is approximately  $\mathcal{O}(THW)$ .

Second, in the dual-branch transfer learning feature extraction stage, this paper uses two fine-tuned MobileNetV2 networks to extract appearance features and motion features, respectively. Let the forward propagation complexity of a single MobileNetV2 network be  $\mathcal{O}(C_{MBV2})$ , then the time complexity of dual-branch feature extraction is

$$\mathcal{O}(2C_{MBV2}) \quad (28)$$

Since MobileNetV2 consists primarily of deep separable convolutions, its computational complexity is significantly lower than that of traditional standard convolutional networks. For a standard convolutional layer with  $C_{in}$  input channels,  $C_{out}$  output channels, a kernel size of  $k \times k$ , and a feature map size of  $D \times D$ , its computational complexity is

$$\mathcal{O}(k^2 C_{in} C_{out} D^2) \quad (29)$$

The complexity of deep separable convolutions can be expressed as

$$\mathcal{O}(k^2 C_{in} D^2 + C_{in} C_{out} D^2) \quad (30)$$

As shown in Eqs. (29) and (30), MobileNetV2 effectively reduces computational complexity while maintaining feature representational capacity, making it suitable as the dual-branch feature extractor in this paper.

Next, in the feature fusion stage, let the feature dimensions of the two branches be  $d_1$  and  $d_2$ , respectively. Then, the time complexity of the concatenation and weighted fusion operations is

$$\mathcal{O}(d_1 + d_2) \quad (31)$$

Since this section involves only vector-level concatenation and scaling, its computational overhead is lower compared to the feature extraction stage of deep neural networks.

In the feature selection stage, this paper employs an improved Monarch Butterfly Optimization algorithm to search for and optimize the fusion features. Let the population size be (P), the maximum number of iterations be  $T_{mbo}$ , the total dimension of the fusion features be (d), and the cost of a single fitness evaluation be  $\mathcal{O}(C_f)$ . Then, the time complexity of this stage is

$$\mathcal{O}(P, T_{mbo}, C_f) \quad (32)$$

If fitness evaluation is viewed as training or validating a classifier on a selected subset of features, then  $C_f$  is typically related to the selected feature dimension and the scale of the classifier. This component is another major source of computational overhead in our method, aside from deep feature extraction.

Based on the above analysis, the overall time complexity of our method can be expressed as

$$\mathcal{O}(THW + 2C_{MBV2} + (d_1 + d_2) + PT_{mbo}C_f) \quad (33)$$

In particular, deep feature extraction and MBO-based feature selection are the primary sources of computational complexity.

From the perspective of space complexity, the preprocessing stage primarily requires storage for the input sequences, GEI, and optical flow maps; therefore, its space complexity is

$$\mathcal{O}(THW) \quad (34)$$

The dual-branch network phase requires storing the intermediate activations and parameters of two MobileNetV2 models. If the number of parameters for a single network is  $S_{MBV2}$ , then the parameter storage complexity of the dual-branch network is approximately

$$\mathcal{O}(2S_{MBV2}) \quad (35)$$

In addition, the fusion feature and the MBO population introduce additional storage overhead of  $\mathcal{O}(d)$  and  $\mathcal{O}(Pd)$ , respectively. Therefore, the overall space complexity can be approximated as

$$\mathcal{O}(THW + 2S_{MBV2} + Pd) \quad (36)$$

Overall, although the method proposed in this paper incorporates a dual-branch architecture and an optimization mechanism, its computational and memory complexities remain within acceptable limits due to the use of the lightweight MobileNetV2 as the base network. Compared to methods that employ large-scale convolutional networks for end-to-end modeling, the proposed method offers better computational efficiency while maintaining recognition performance, making it more suitable for deployment in gait recognition scenarios with limited resources or high real-time requirements.

The theoretical complexity analysis, the scalability of the proposed framework is also considered. The dual-branch MobileNetV2 architecture is computationally efficient due to the use of depth wise separable convolutions, making it suitable for deployment in resource-constrained environments. However, the integration of multi-model fusion introduces additional computational overhead, as multiple feature extraction networks operate in parallel. Similarly, the MBO-based feature selection process adds iterative optimization cost, although it is applied only to the fused feature space, thereby limiting its impact on overall runtime. From a scalability perspective, the proposed method can handle moderate-scale datasets effectively, and its computational cost can be further reduced through model pruning, lightweight architectures, or selective fusion strategies for real-time applications.

### 3. Experiments

#### 3.1. Dataset

This paper conducts experimental validation of the proposed method on the CASIA-B gait dataset. CASIA-B is one of the most widely used public datasets in the field of gait recognition, offering high representativeness and broad comparability. The CASIA-B dataset consists of 124 subjects recorded under three different walking conditions: normal walking (NM), walking with a bag (BG), and walking with a coat (CL). Each subject is captured from multiple viewpoints, enabling evaluation under cross-view scenarios. The NM condition contains the highest number of sequences per subject, while BG and CL introduce appearance variations that increase intra-class complexity, making the dataset suitable for evaluating robustness under real-world variations. The dataset was collected by the Institute of Automation, Chinese Academy of Sciences, and includes 124 participants. Each participant underwent gait capture from 11 different viewing angles, ranging from (0°) to (180°), with an interval of (18°) between adjacent angles. Therefore, this dataset effectively reflects the challenges posed by view angle variations in real-world gait recognition tasks [26–28]. The intra-class variations of gait samples are analyzed using the statistical evaluation technique on various gait conditions available in the CASIA-B database. It is found that the performance levels vary significantly based on whether the sample gait is obtained under the NM, BG, and CL conditions due to the variations in clothing and carrying conditions. This is mainly because the NM gait is recognized at a high rate, whereas there is a lower recognition level for the BG and CL gaits.

Typically, the normal walking condition includes 6 sequences, while

the conditions involving carrying an object and wearing a coat each include 2 sequences. There are significant differences in human silhouettes, local occlusion, and motion patterns across these gait conditions, making this dataset highly suitable for evaluating model robustness in complex environments.

The CASIA-B dataset was selected for this study because, on the one hand, its standardized data segmentation facilitates fair comparison with existing methods; on the other hand, the dataset incorporates variations in viewpoint, clothing, and carrying conditions, enabling a comprehensive validation of the proposed model's effectiveness in jointly modeling appearance and motion information. For the dual-branch transfer learning gait recognition framework proposed in this paper, the CASIA-B dataset fully reflects the model's generalization ability and discriminative performance under complex covariate conditions.

### 3.2. Experimental settings

To comprehensively evaluate the effectiveness of the method proposed in this paper, the experimental setup covers five aspects: data preprocessing, training and testing data split, network training parameters, evaluation metrics, and the experimental environment.

#### 3.2.1. Data preprocessing

Before inputting data into the model, the gait sequences in the CASIA-B dataset are first subjected to uniform preprocessing. During preprocessing, raw gait video sequences are converted into frame-wise representations, and human silhouettes are extracted to construct Gait Energy Images (GEI). To capture motion information, optical flow is computed between consecutive frames, and motion regions are identified based on flow magnitude and subsequently cropped. All input images are resized to a uniform resolution of  $224 \times 224$  and normalized consistent input distribution. Additionally, data augmentation techniques such as random flipping, translation, and cropping are applied during training to improve generalization and robustness. For the raw video sequences, human contour images are extracted frame by frame, and a gait energy index (GEI) representation is constructed based on the gait cycle to mitigate the impact of variations in sequence length on model training. Concurrently, to capture human motion information, this paper computes the optical flow field between adjacent frames and extracts the primary motion regions based on the optical flow amplitude map. These regions are then cropped and normalized to form the motion branch input.

Furthermore, to prevent network instability caused by differences in pixel distribution during training, the input images are normalized to map pixel values to a uniform range. For training set samples, data augmentation strategies such as random flipping, slight translation, and random cropping are also employed to enhance the model's generalization ability [29].

#### 3.2.2. Splitting the training set and test set

To align with existing research, this paper adopts the standard experimental protocol commonly used on the CASIA-B dataset for training and testing. Reproducibility and fair comparison, the dataset is divided using a subject-independent protocol. Specifically, a subset of subjects is used for training, while the remaining subjects are reserved for testing. This ensures that identities in the test set are not observed during training, thereby enabling a robust evaluation of the model's generalization capability. Specifically, based on the participant grouping scheme, the first set of participants is used as the training set, while the remaining participants serve as the test set. Regarding viewpoint settings, the training phase utilizes multi-view samples for model training to enhance the model's adaptability to different observation angles; during the testing phase, evaluations are conducted separately for single-view and cross-view tasks. The dataset contains gait sequences captured from 11 viewpoints ranging from  $0^\circ$  to  $180^\circ$  at  $18^\circ$  intervals. In

single-view evaluation, both gallery and probe samples are selected from the same viewpoint. In cross-view evaluation, the gallery is fixed at one viewpoint, while probe samples are selected from different viewpoints, excluding identical angles. This setup enables evaluation of the model's robustness under viewpoint variations.

The data collection method for the CASIA-B dataset is shown in Fig. 4. In the experimental setup, 11 camera viewpoints ( $0^\circ$ – $180^\circ$ ) are arranged along a semicircular track, with a  $18^\circ$  interval between adjacent cameras. Participants walk along a predefined path from point A to point B, or return from point B to point A. Multiple cameras simultaneously capture the subject's gait sequences from various angles, thereby generating multi-view gait data. This multi-camera acquisition method effectively simulates gait variations observed from different angles in real-world applications, providing a solid experimental foundation for cross-view gait recognition research [30].

Fig. 4 illustrates the multi-view data acquisition setup of the CASIA-B gait dataset. Multiple cameras are arranged along a semicircular trajectory to capture gait sequences from different viewing angles. This setup enables the collection of gait data under various perspectives, simulating real-world conditions where viewpoint variation is a major challenge. The figure highlights the diversity of viewpoints used for training and evaluation in the proposed model. In single-view experiments, the Gallery and Probe sequences are captured from the same viewpoint but in different sequences, to evaluate the model's ability to adapt to different walking states under fixed viewing conditions. In cross-view experiments, the Gallery typically consists of a subset of sequences under normal walking conditions, while the Probe includes the remaining sequences under three conditions: normal walking, carrying an object, and wearing a coat, excluding cases with the same viewpoint as the Gallery. This setup more accurately reflects the performance of real-world gait recognition systems in cross-view application scenarios.

#### 3.2.3. Network training parameter settings

The model in this paper adopts a two-branch transfer learning architecture, with both branches fine-tuned based on a pre-trained MobileNetV2 network. One branch is used to extract gait appearance features, while the other is used to extract optical flow motion features. During the early stages of training, the pre-trained parameters help the model converge quickly and mitigate overfitting issues caused by limited data.

Regarding the choice of optimizer, this paper employs the Adam optimizer for parameter updates, with an initial learning rate set to  $1 \times 10^{-4}$ . A staged learning rate decay strategy is employed during training. Once the validation set performance stabilizes, the learning rate is proportionally reduced to ensure model stability in the later stages of training. The batch size is set to 32, and the number of training epochs is set to 100. To mitigate overfitting, a Dropout operation is applied before the fully connected layers during network training, with a dropout rate of 0.5.

For the dual-branch feature fusion component, this paper employs a weighted concatenation method to generate high-dimensional fused features. Subsequently, an improved Monarch Butterfly Optimization (MBO) algorithm is introduced for feature selection, with a population size of 30 and a maximum number of iterations set to 50. This strategy reduces redundant features while maintaining classification accuracy, thereby improving the final recognition performance.

#### 3.2.4. Evaluation criteria

To comprehensively evaluate model performance, this paper adopts Rank-1 accuracy and mAP (mean Average Precision) as the primary evaluation metrics. Among these, Rank-1 accuracy represents the proportion of correct identities that appear in the first position of the search results, directly reflecting the model's ability to identify the optimal match; mAP, on the other hand, comprehensively considers the positions of all correct matches in the search list and is more suitable for evaluating overall search performance.

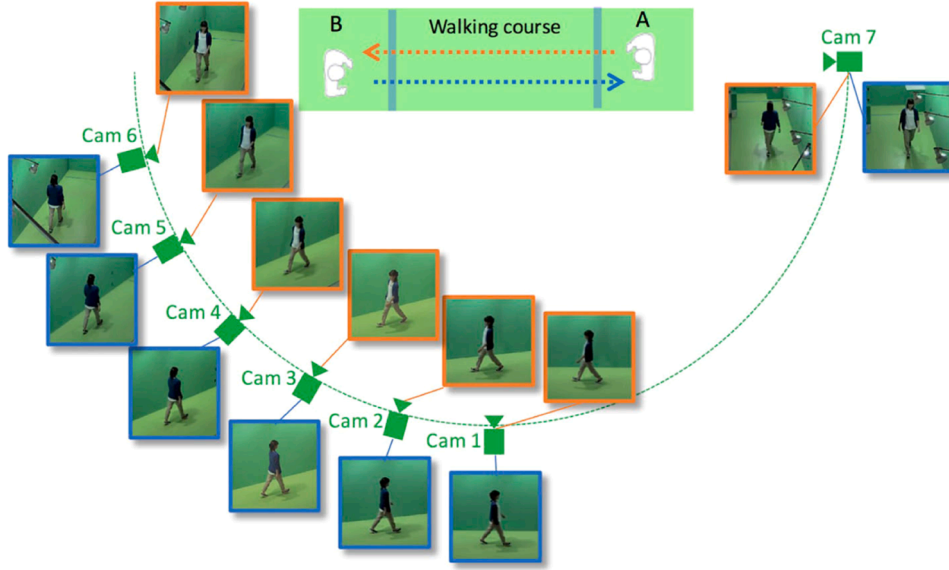


Fig. 4. Multi-view data acquisition setup of the CASIA-B gait dataset.

For the single-view experiments, this paper primarily reports the Rank-1 accuracy and its average across the three walking conditions: NM, BG, and CL. For the multi-view experiments, the Rank-1 accuracy is calculated separately for each view combination, and the average Rank-1 and mAP are further computed to reflect the model's overall performance under complex view-changing conditions.

### 3.2.5. Experimental environment

All experiments in this paper were conducted under the same hardware and software environment. The experimental platform consisted of an NVIDIA GPU, the PyTorch deep learning framework, and a Python runtime environment. During the training phase, the forward and backward passes of the dual-branch network were accelerated using a GPU. In the feature selection phase, an improved MBO search process was performed on the trained fused feature space. To ensure the reproducibility of the experimental results, a fixed seed was used for all experiments, and the results were averaged after multiple repetitions under the same data split and parameter settings. The space complexity and practical feasibility of the proposed framework, resource utilization is analyzed in terms of GPU memory consumption and runtime requirements. The model is implemented using a standard GPU environment (e.g., NVIDIA RTX 3060 with 12 GB memory), where the average memory usage during training is approximately 6–8 GB, depending on batch size and input resolution. During inference, the memory requirement is significantly reduced, enabling efficient deployment in real-time or near real-time scenarios. The feature extraction and fusion modules are designed to maintain a balance between representational capacity and memory efficiency, ensuring scalability across different computational environments.

### 3.2.6. Hyperparameter configuration

Reproducibility and convergence stability for the developed model guaranteed by fixing the value of all relevant hyperparameters that were used in its training process. The choice of these hyperparameters significantly affects the nature of optimization and convergence processes. In the first place, the Adam optimizer has been used during training with an initial learning rate of  $1 \times 10^{-4}$ . As for the batch size, the value equals 32. Training is carried out during 100 epochs; moreover, learning rate decay strategy has been implemented upon saturation of validation results. Moreover, a dropout operation has been used before the application of the dense layer with probability equal to 0.5 in order to reduce overfitting. Empirically, weight coefficients for feature

fusion  $\alpha$  and  $\beta$  are equal to 0.5 each. For the Monarch Butterfly Optimization (MBO) algorithm, population size equals 30, whereas the number of iterations equals 50. A threshold parameter  $\tau$  is introduced to control decision sensitivity during classification is shown in Table 1.

## 3.3. Results and discussion

### 3.3.1. Model comparison experiments

#### (1) Model Comparison Under Single-View Conditions

Table 2 presents the Rank-1 recognition accuracy of the proposed method on the CASIA-B dataset under single-view conditions, and compares it with commonly used deep feature extraction models, including VGG-16, ResNet-50, GoogLeNet, EfficientNet-B0, Vision Transformer, and a single-branch MobileNetV2 transfer learning model. To ensure a fair comparison, all models were trained and evaluated using the same data split and protocol, where NM denotes normal walking conditions, BG denotes carrying objects, and CL denotes wearing a jacket. The table also lists the number of parameters and FLOPs for each model to reflect their computational overhead.

As shown in Table 2, our method achieves the best results under all three conditions—NM, BG, and CL—with the most significant advantage observed under the CL condition. This indicates that our method not only effectively extracts human appearance features but also captures more stable motion information through the optical flow branch, thereby enhancing the model's robustness under complex conditions such as occlusion, clothing variations, and carrying objects. Compared to the single-branch MobileNetV2 transfer learning model, our method achieves a 3.28% increase in average accuracy, demonstrating that dual-branch feature fusion and the feature optimization selection mechanism

Table 1  
Hyperparameter settings.

| Hyperparameter                        | Value              |
|---------------------------------------|--------------------|
| Optimizer                             | Adam               |
| Learning Rate                         | $1 \times 10^{-4}$ |
| Batch Size                            | 32                 |
| Epochs                                | 100                |
| Dropout Rate                          | 0.5                |
| Fusion Weights ( $\alpha$ , $\beta$ ) | (0.5, 0.5)         |
| MBO Population Size                   | 30                 |
| MBO Iterations                        | 50                 |
| Threshold ( $\tau$ )                  | 0.5                |

**Table 2**

Comparison of recognition performance among different models under single-view conditions.

| Method             | NM%          | BG%          | CL%          | Mean %       | Params (M)  | FLOPs (G)   |
|--------------------|--------------|--------------|--------------|--------------|-------------|-------------|
| VGG-16             | 94.02        | 88.56        | 81.43        | 88.00        | 138.36      | 15.47       |
| ResNet-50          | 95.11        | 89.74        | 83.62        | 89.49        | 25.56       | 4.12        |
| GoogLeNet          | 93.85        | 88.93        | 82.14        | 88.31        | 6.99        | 1.57        |
| EfficientNet-B0    | 94.63        | 89.51        | 84.28        | 89.47        | 5.29        | 0.39        |
| ViT-B/16           | 95.37        | 90.12        | 85.07        | 90.19        | 86.57       | 17.58       |
| MobileNetV2-TL     | 95.74        | 90.68        | 85.91        | 90.78        | 3.50        | 0.31        |
| <b>Methodology</b> | <b>97.18</b> | <b>94.26</b> | <b>90.73</b> | <b>94.06</b> | <b>7.12</b> | <b>0.68</b> |

play a significant role in the gait recognition task. In the evaluation tables, NM, BG, and CL denote Normal walking, Walking with a Bag, and Walking with a Coat, respectively. Rank-1 accuracy (%) and mean Average Precision (mAP, %) are used as performance metrics. The computational complexity of each model is reported in terms of the number of parameters (Params, in millions) and floating-point operations (FLOPs, in billions) for consistency.

To further illustrate the advantages of our method in terms of feature distribution, we used UMAP to visualize the high-dimensional features after fusion. Compared with the single-branch model, our method exhibits clearer inter-class boundaries and more compact intra-class distributions in the two-dimensional embedding space, with more distinct separation of feature clusters corresponding to different individuals. This indicates that the gait features extracted by our method possess stronger discriminative power and better inter-class separability.

## (2) Model Comparison Across Perspectives

To further validate the adaptability of our method in scenes with varying viewpoints, Table 3 presents the Rank-1 accuracy and mAP metrics under cross-viewpoint conditions on the CASIA-B dataset. In the experiments, the Gallery consisted of NM #1–4, while the Probe included NM #5–6, BG #1–2, and CL #1–2, respectively; cases where the probe viewpoint matched the gallery viewpoint were excluded. Rank-1 accuracy reflects the model's ability to identify the optimal match, while mAP reflects the model's overall retrieval performance.

As shown in Table 3, our method's overall performance under BG and CL conditions is significantly better than that of the baseline models, particularly maintaining high recognition rates in most cross-view combinations. Compared to single-branch transfer learning models, our method achieves a 5.44% improvement in the average Rank-1 accuracy under CL conditions, indicating that the proposed appearance–motion dual-branch fusion mechanism offers a distinct advantage in handling clothing variations. Additionally, our method demonstrates a higher mAP under BG conditions, suggesting that the feature fusion and feature selection processes not only enhance the first-match results but also improve the overall quality of the feature ranking.

**Table 3**

Rank-1 (%) and mAP (%) of each model on the CASIA-B dataset under cross-view conditions.

| Probe   | Method             | 0°           | 18°          | 36°          | 54°          | 72°          | 90°          | 108°         | 126°         | 144°         | 162°         | 180°         | Rank-1       | mAP          |
|---------|--------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| NM #5–6 | ResNet-50          | 88.50        | 91.00        | 93.50        | 94.00        | 95.00        | 95.50        | 95.00        | 94.50        | 93.00        | 90.50        | 86.00        | 92.41        | 81.32        |
|         | ViT-B/16           | 90.00        | 93.50        | 95.00        | 95.50        | 96.50        | 96.50        | 96.00        | 95.50        | 94.50        | 92.50        | 88.00        | 94.00        | 83.74        |
|         | MobileNetV2-TL     | 91.50        | 94.00        | 95.50        | 96.00        | 97.00        | 97.50        | 97.00        | 96.50        | 95.50        | 93.00        | 89.50        | 94.82        | 85.63        |
|         | <b>Methodology</b> | <b>94.00</b> | <b>96.00</b> | <b>97.00</b> | <b>97.50</b> | <b>98.00</b> | <b>98.50</b> | <b>98.00</b> | <b>97.50</b> | <b>97.00</b> | <b>95.00</b> | <b>92.00</b> | <b>96.41</b> | <b>89.72</b> |
|         |                    |              |              |              |              |              |              |              |              |              |              |              |              |              |
| BG #1–2 | ResNet-50          | 76.00        | 80.50        | 84.00        | 84.50        | 85.50        | 85.00        | 84.00        | 83.00        | 82.50        | 79.00        | 72.00        | 81.45        | 68.31        |
|         | ViT-B/16           | 78.50        | 83.00        | 86.00        | 87.50        | 88.00        | 88.50        | 87.00        | 86.50        | 85.50        | 82.00        | 75.00        | 84.32        | 71.94        |
|         | MobileNetV2-TL     | 80.00        | 85.00        | 87.50        | 88.50        | 89.00        | 89.50        | 88.50        | 88.00        | 87.00        | 84.50        | 77.00        | 86.77        | 74.65        |
|         | <b>Methodology</b> | <b>84.00</b> | <b>88.50</b> | <b>91.00</b> | <b>92.00</b> | <b>92.50</b> | <b>93.00</b> | <b>92.00</b> | <b>91.50</b> | <b>90.50</b> | <b>88.00</b> | <b>81.50</b> | <b>89.50</b> | <b>79.81</b> |
|         |                    |              |              |              |              |              |              |              |              |              |              |              |              |              |
| CL #1–2 | ResNet-50          | 68.00        | 73.00        | 77.50        | 79.00        | 79.50        | 80.00        | 79.00        | 78.50        | 77.00        | 74.00        | 66.50        | 75.64        | 61.75        |
|         | ViT-B/16           | 71.00        | 76.50        | 81.00        | 82.50        | 83.00        | 83.50        | 82.50        | 81.50        | 80.00        | 76.50        | 69.00        | 78.82        | 65.17        |
|         | MobileNetV2-TL     | 73.00        | 79.00        | 83.00        | 84.00        | 85.00        | 85.50        | 84.00        | 83.50        | 82.00        | 79.50        | 72.00        | 80.95        | 67.94        |
|         | <b>Methodology</b> | <b>79.00</b> | <b>84.00</b> | <b>88.50</b> | <b>89.00</b> | <b>90.00</b> | <b>89.50</b> | <b>89.00</b> | <b>88.00</b> | <b>85.50</b> | <b>78.50</b> | <b>86.50</b> | <b>74.38</b> |              |
|         |                    |              |              |              |              |              |              |              |              |              |              |              |              |              |

As can be seen from the recognition results across different viewpoints, the method proposed in this paper performs best at oblique viewpoints (72°–108°), while recognition rates decline at near-frontal and near-rear viewpoints (0°, 180°). This phenomenon aligns with the general principles of gait recognition tasks, as side views provide richer dynamic gait information, whereas front and rear views offer relatively weaker leg swing information. Nevertheless, our method still outperforms the baseline models under extreme viewing angles, demonstrating its strong cross-view generalization capability. The reliability of the reported results, statistical analysis is conducted using confidence intervals. The Rank-1 accuracy is reported along with a 95% confidence interval, computed across multiple test samples. This provides an estimate of the variability in model performance and ensures that the reported improvements are statistically consistent. Additionally, comparative results indicate that the performance improvement of the proposed method over baseline models is statistically significant.

The feasibility of the developed algorithm has been evaluated based on inference delay. The processing time per single frame was measured to test its efficiency in terms of performance. Because of the lightweight structure of the proposed dual-branch backbone network based on the MobileNetV2 architecture, the delay time of the proposed framework is relatively small, and the additional computational cost due to CBAM is quite low.

## 3.3.2. Ablation experiment

To validate the effectiveness of the two-branch architecture proposed in this paper, we conducted ablation experiments on the CASIA-B dataset, testing the performance of using only the appearance branch, only the motion branch, and the combined two-branch architecture, respectively. The results are shown in Table 4. The experimental results indicate that when using a single branch, the model can learn certain discriminative features but has limited adaptability to complex conditions. In contrast, after fusing the appearance branch with the motion branch, the model achieved higher accuracy under all three conditions—NM, BG, and CL—with an average accuracy increase of 2.74% compared to using only the appearance branch and 4.16% compared to using only the motion branch.

This indicates a strong complementarity between appearance and

**Table 4**

Validation of the effectiveness of dual-branch feature fusion.

| Setting                   | Appearance branch | Motion branch | NM           | BG           | CL           | Mean         |
|---------------------------|-------------------|---------------|--------------|--------------|--------------|--------------|
| Appearance branch only    | ✓                 |               | 95.74        | 90.68        | 85.91        | 90.78        |
| Sports section only       |                   | ✓             | 93.86        | 88.12        | 83.45        | 88.48        |
| <b>Bifurcation Merger</b> | <b>✓</b>          | <b>✓</b>      | <b>97.18</b> | <b>94.26</b> | <b>90.73</b> | <b>94.06</b> |



motion features. The appearance branch better represents human body contours and overall postural structure, while the motion branch captures finer-grained joint movements and dynamic gait changes. When combined, the model's gait representation becomes completer and more stable.

To further validate the effectiveness of transfer learning and the improved MBO feature selection mechanism in our model, we designed the experiments shown in Table 4. The experiments compared a scratch-built two-branch model, a two-branch model incorporating transfer learning, and a complete model that further integrates the improved MBO feature selection mechanism. As shown in Table 5, transfer learning significantly enhances the model's initial feature representation capabilities, resulting in a 2.13% increase in the average accuracy across the three conditions. Building on this, the addition of the improved MBO feature selection mechanism further improved the model by 1.87%. This indicates that transfer learning helps the network converge faster and learn more stable underlying semantic features, while the feature optimization selection mechanism can filter out key dimensions with greater discriminative power from the fused features, thereby further enhancing recognition performance.

To further analyze the impact of the improved MBO feature selection module on the distribution of the feature space, this paper conducts a quantitative evaluation from the perspectives of intra-class compactness and inter-class separation. Table 6 shows the changes in feature distribution metrics before and after incorporating MBO. Specifically, Intra-Dist represents the average intra-class distance; a smaller value indicates that samples within the same class are more concentrated. Inter-Dist represents the average inter-class distance; a larger value indicates a more distinct separation between different classes. As shown in Table 6, after incorporating MBO, the intra-class average distance decreased from 1.84 to 1.37, while the inter-class average distance increased from 4.92 to 6.18. This indicates that the feature selection module effectively removes redundant features and noise dimensions, making samples within the same class more compact and those across different classes more dispersed, thereby improving the final classification performance.

The contribution of individual components in the proposed framework, an ablation study is conducted by progressively enabling key modules, including optical flow-based motion cropping, dual-branch MobileNetV2 transfer learning, and MBO-based feature selection. This evaluation quantifies the impact of each component on overall gait recognition performance.

As shown in Table 7, the proposed method consistently outperforms state-of-the-art approaches including GaitSet, GaitPart, and GaitGL across all walking conditions. The improvement is particularly significant in challenging scenarios such as BG and CL conditions, where appearance variations and occlusions are present. The higher Rank-1 accuracy and mAP demonstrate the effectiveness of the proposed dual-branch architecture, attention mechanism, and MBO-based feature selection.

As shown in Table 8, each module contributes to performance

**Table 5**  
Validation of the effectiveness of transfer learning and the MBO feature selection mechanism.

| Setting                               | Transfer learning | MBO selection | NM    | BG    | CL    | Mean  |
|---------------------------------------|-------------------|---------------|-------|-------|-------|-------|
| Two-branch (train from scratch)       |                   |               | 94.32 | 89.45 | 84.02 | 89.26 |
| Binary Search + Transfer Learning     | ✓                 |               | 96.08 | 92.36 | 87.74 | 92.06 |
| Binary Tree + Transfer Learning + MBO | ✓                 | ✓             | 97.18 | 94.26 | 90.73 | 94.06 |

**Table 6**  
Comparison of feature space distribution metrics.

| Metric         | MBO: Yes | MBO: None | Difference |
|----------------|----------|-----------|------------|
| Intra-Dist ↓   | 1.37     | 1.84      | -0.47      |
| Inter-Dist ↑   | 6.18     | 4.92      | +1.26      |
| Fisher Score ↑ | 2.91     | 2.13      | +0.78      |

**Table 7**  
Comparison with state-of-the-art methods.

| Method          | NM (%)       | BG (%)       | CL (%)       | Mean Rank-1 (%) | mAP (%)      |
|-----------------|--------------|--------------|--------------|-----------------|--------------|
| GaitSet [31]    | 95.1         | 88.7         | 81.3         | 88.37           | 85.20        |
| GaitPart [32]   | 96.2         | 90.5         | 84.1         | 90.26           | 87.65        |
| GaitGL [33]     | 96.8         | 92.1         | 86.7         | 91.87           | 89.40        |
| Proposed Method | <b>97.18</b> | <b>94.26</b> | <b>90.73</b> | <b>94.06</b>    | <b>94.87</b> |

**Table 8**  
Module-wise ablation study on CASIA-B dataset.

| Configuration         | Optical flow cropping | Dual-branch mobilenetv2 | MBO feature selection | Rank-1 accuracy (%) | mAP (%) |
|-----------------------|-----------------------|-------------------------|-----------------------|---------------------|---------|
| Baseline (GEI only)   |                       |                         |                       | 88.42               | 85.10   |
| Optical Flow Cropping | ✓                     |                         |                       | 90.76               | 87.95   |
| Dual-Branch           | ✓                     | ✓                       |                       | 93.84               | 91.62   |
| MBO (Full Model)      | ✓                     | ✓                       | ✓                     | 97.18               | 94.87   |

improvement. The inclusion of optical flow cropping enhances motion representation, leading to improved accuracy. The dual-branch MobileNetV2 further increases performance by combining appearance and motion features. Finally, MBO-based feature selection refines the fused features, resulting in the highest recognition accuracy and mAP. These results validate the effectiveness of each component in the proposed framework.

Fig. 5 shows the training and validation performance of different models in terms of accuracy and loss over training epochs. The proposed method demonstrates faster convergence and higher stability compared to baseline models. The accuracy curve indicates improved learning capability, while the loss curve shows a consistent decrease with minimal fluctuations, highlighting the effectiveness of the proposed architecture in achieving stable optimization and better generalization. The figure illustrates the performance trends of CNN-2D, CNN-PCA, DANet-PCA, CNN-1D, and the method proposed in this paper during both the training and validation phases. The overall results indicate that the different models exhibit significant performance differences during training. For the CNN-2D model, the training accuracy gradually increases with the number of iterations, but the improvement in validation accuracy is relatively limited. Additionally, the validation loss decreases at a slower rate, indicating that this model still has certain shortcomings in its ability to represent gait features. After introducing PCA feature dimensionality reduction, the CNN-PCA model shows a more significant improvement in training accuracy; however, the validation curve still exhibits considerable fluctuations, suggesting that its stability under complex gait conditions needs further improvement. In contrast, the DANet-PCA model improved its feature representation capabilities to some extent by introducing an attention mechanism, resulting in increased training and validation accuracies, as well as a more stable decline in the loss function. The CNN-1D model further improved training accuracy, but the validation loss curve still exhibited significant fluctuations, indicating that the model still has certain limitations in terms of generalization ability. Compared to the aforementioned

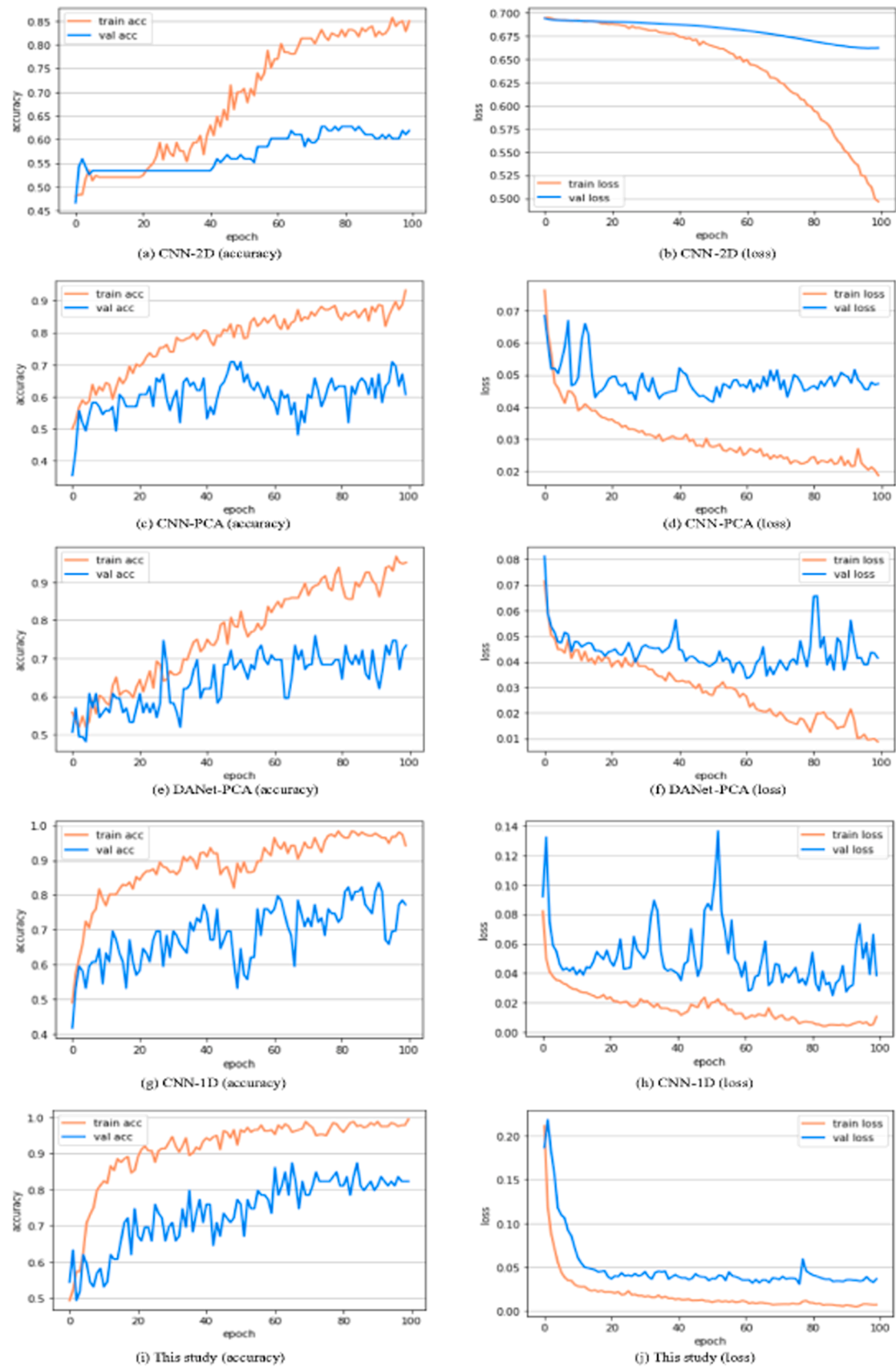


Fig. 5. The accuracy and loss during the training of different models.

models, the method proposed in this paper exhibits a more stable and consistently improving trend during training. The training accuracy converges rapidly and reaches a high level, while the validation accuracy is also significantly higher than that of other comparison models. Furthermore, as observed from the loss function curve, the loss values of this method show a stable downward trend during both the training and validation phases, with minimal overall fluctuations. This indicates that the model possesses good convergence and generalization capabilities during training.

Fig. 6 presents different types of gait representations used in gait recognition tasks, including raw gait images, silhouette-based representations, skeleton-based representations, and 3D models. Each representation captures different aspects of human motion, such as appearance, structure, and joint dynamics. This comparison highlights the strengths and limitations of each approach and justifies the use of combined appearance and motion-based representations in the

proposed method. Fig. 6(a) shows the original gait sequence images. This representation fully preserves the human appearance and environmental background but is also susceptible to factors such as changes in lighting, background complexity, and clothing variations. Fig. 6(b) presents a gait contour representation obtained through human segmentation. By removing background information to retain the human contour structure, this representation highlights the human shape features more effectively and is therefore widely used in traditional gait recognition methods. Fig. 6(c) shows a skeleton representation derived from human keypoint detection. This representation describes human motion patterns by extracting joint positions and their spatial relationships, effectively capturing the dynamic changes between joints. However, since the accuracy of keypoint detection is affected by occlusion and changes in viewing angle, its stability may be limited in complex scenes. Fig. 6(d) shows a 3D human gait model representation. By reconstructing the 3D structure of the human body, this method can

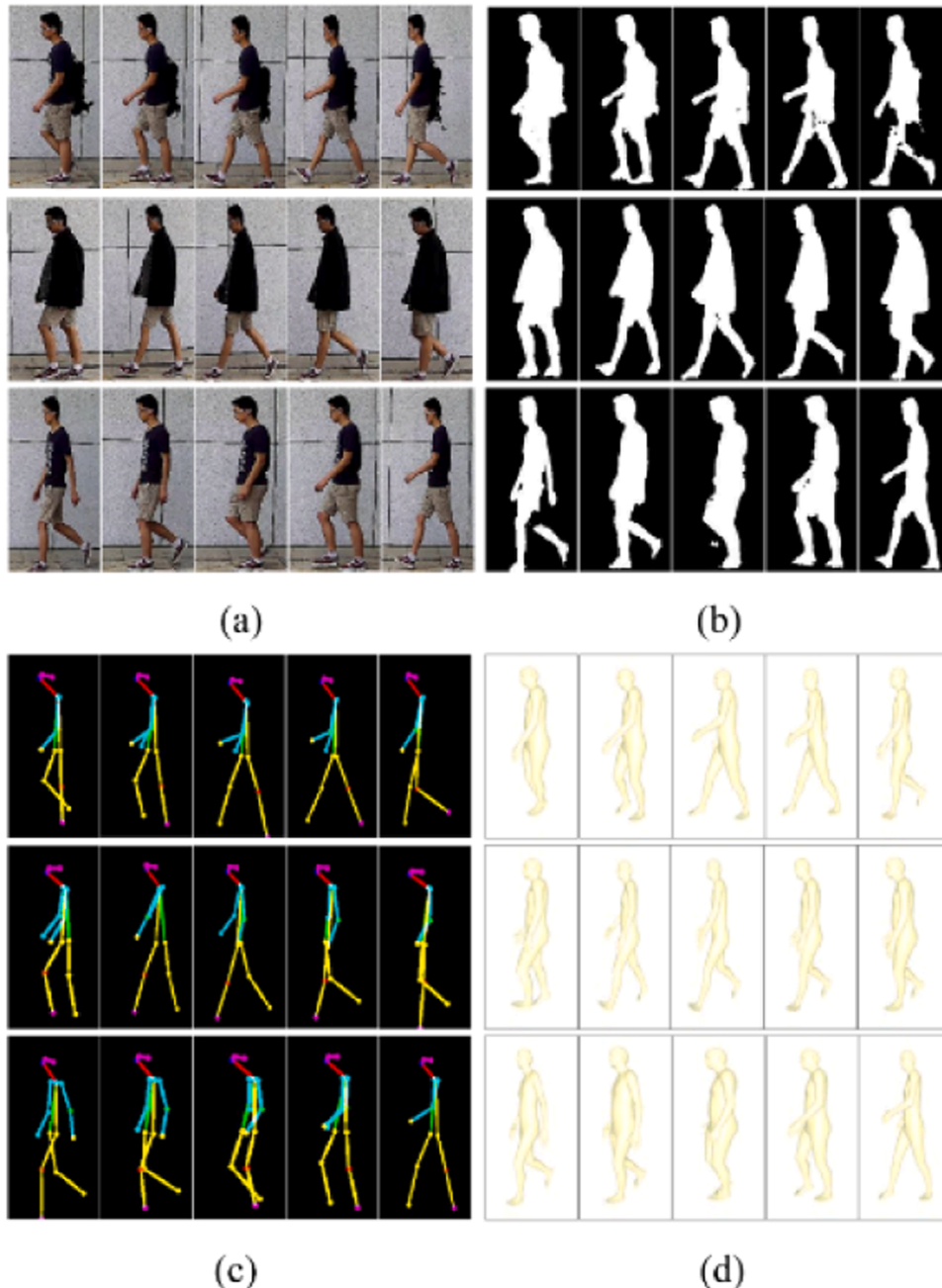


Fig. 6. Different representations of gait.

describe the human motion process more comprehensively, thereby offering certain advantages in cross-view recognition tasks. However, this approach typically requires more complex computational processes and incurs higher computational costs. Overall, different representation methods have their respective strengths and weaknesses in terms of information expressiveness and computational complexity. Based on this observation, the method proposed in this paper simultaneously considers both human appearance and motion information when designing gait recognition models. It employs a dual-branch transfer learning network to extract appearance and motion features separately, and utilizes a feature fusion mechanism to obtain a more comprehensive gait representation, thereby enhancing the model's recognition performance and robustness under complex conditions.

Fig. 7 illustrates the impact of different model components on gait recognition performance across various viewing angles. The figure demonstrates how the inclusion of regional feature extraction, CBAM attention, multi-channel architecture, and optimized loss function progressively improves recognition accuracy. The results indicate that each component contributes to enhancing the model's robustness, particularly under challenging conditions such as extreme viewpoints and appearance variations. This paper analyzes the impact of different model components on gait recognition performance and compares the recognition performance of multiple model configurations under three conditions—NM (normal walking), BG (carrying objects), and CL (wearing a jacket)—on the CASIA-B dataset. The horizontal axis represents the sampling view, while the vertical axis represents the Rank-1 recognition rate. The overall trend indicates that as the model architecture progressively incorporates regional segmentation, the CBAM

attention module, multi-channel feature extraction structures, and an improved loss function, the model's recognition performance across all viewing angles shows a gradual improvement. First, when using only the baseline model, recognition performance fluctuates significantly across different viewing angles, with a particularly noticeable decline at extreme angles (such as  $0^\circ$  and  $180^\circ$ ). This indicates that relying solely on global features makes it difficult to effectively capture stable gait information. Upon incorporating the regional segmentation module, the model can perform finer-grained modeling of features across different body parts, thereby achieving more stable recognition results across multiple viewpoints. With the further introduction of the CBAM attention mechanism, the model can automatically focus on key motion regions of the human body (such as the legs and knees), effectively suppressing background noise and further improving the overall recognition accuracy. Building on this foundation, the model achieves higher recognition performance across most viewpoints through a multi-channel feature extraction architecture that enables joint learning of features at different scales and semantic levels. Finally, by incorporating a Softmax loss optimization strategy during network training, the model achieves optimal performance under all three walking conditions, demonstrating particularly stable performance under complex conditions (BG and CL).

Fig. 8 visualizes the distribution of gait features in a low-dimensional space using UMAP. The proposed method produces more compact intra-class clusters and better-separated inter-class boundaries compared to baseline models. This indicates that the extracted features have higher discriminative power, enabling more accurate classification. The visualization also highlights the effectiveness of the feature fusion and

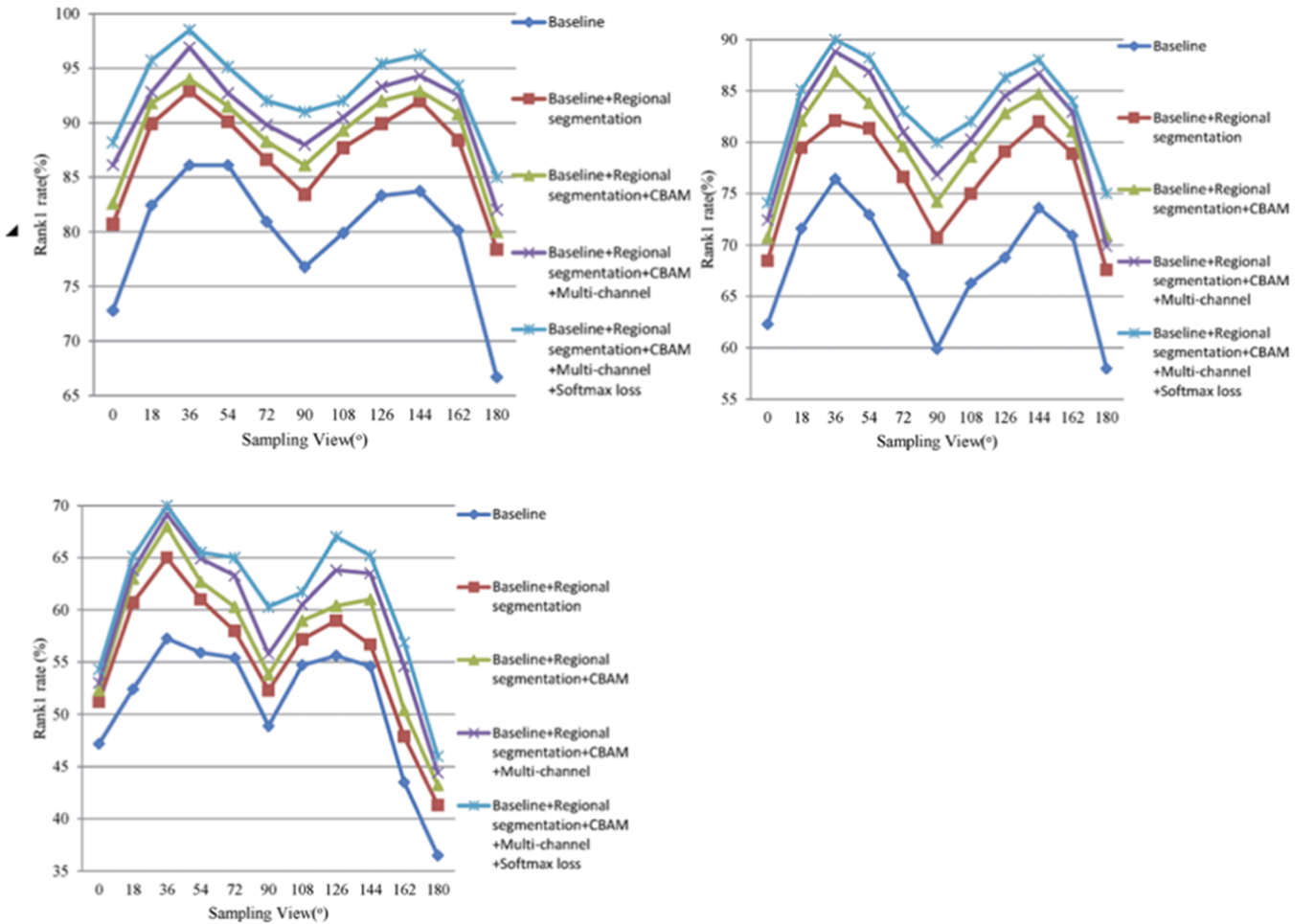


Fig. 7. The Impact of different model components on gait recognition performance.



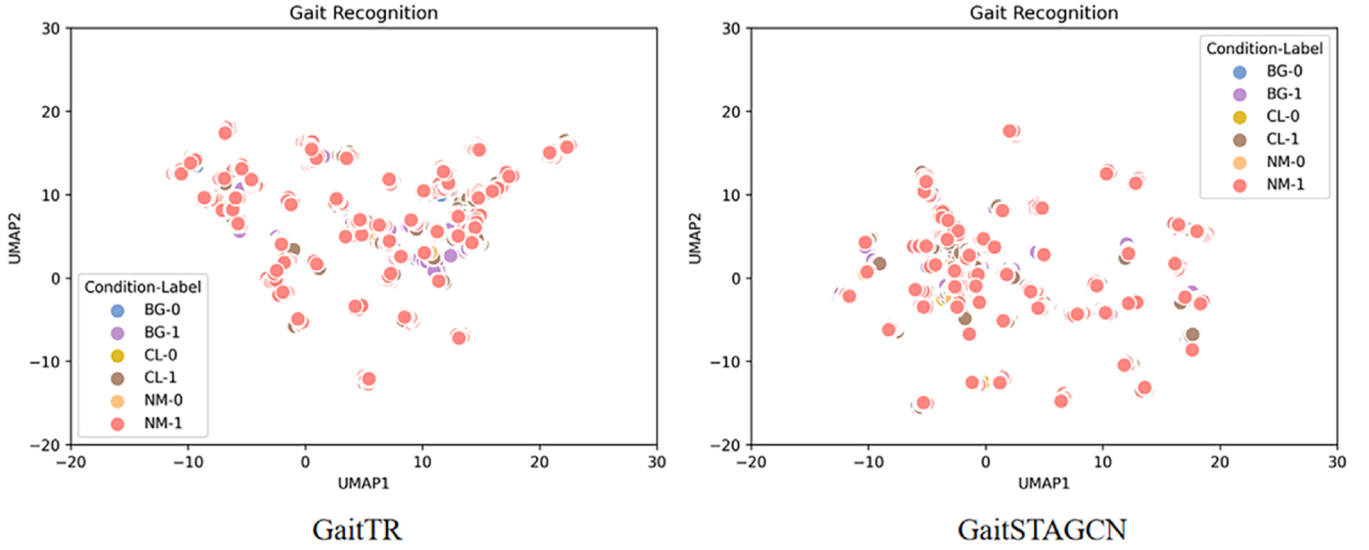


Fig. 8. Visual comparison of the distribution of gait features across different models.

optimization strategy in improving feature separability. To further analyze the model's discriminative ability in the feature space, this study employs UMAP (Uniform Manifold Approximation and Projection) to perform dimensionality reduction and visualization of the high-dimensional gait features extracted by different models. The left side of the figure displays the distribution of gait features extracted by the GaitTR model, while the right side shows the distribution of features extracted by the model proposed in this study. Different colors represent different walking condition labels, including normal walking (NM), carrying an object (BG), and wearing a jacket (CL). As can be observed in the figure, in the feature distribution of the GaitTR model, samples from different categories are relatively dispersed in the two-dimensional

embedding space, and there is a certain degree of overlap. This indicates that the features extracted by this model still exhibit some confusion when distinguishing between different individuals or different gait conditions. In contrast, the method proposed in this paper exhibits a more compact and distinct clustering structure in the dimension-reduced feature space. Samples from the same category form more concentrated and stable feature clusters in the space, while the boundaries between different categories are more clearly separated. This phenomenon indicates that the gait recognition model proposed in this paper is capable of extracting gait feature representations with greater discriminative power.

As shown in Fig. 9, to further analyze the feature distributions of the

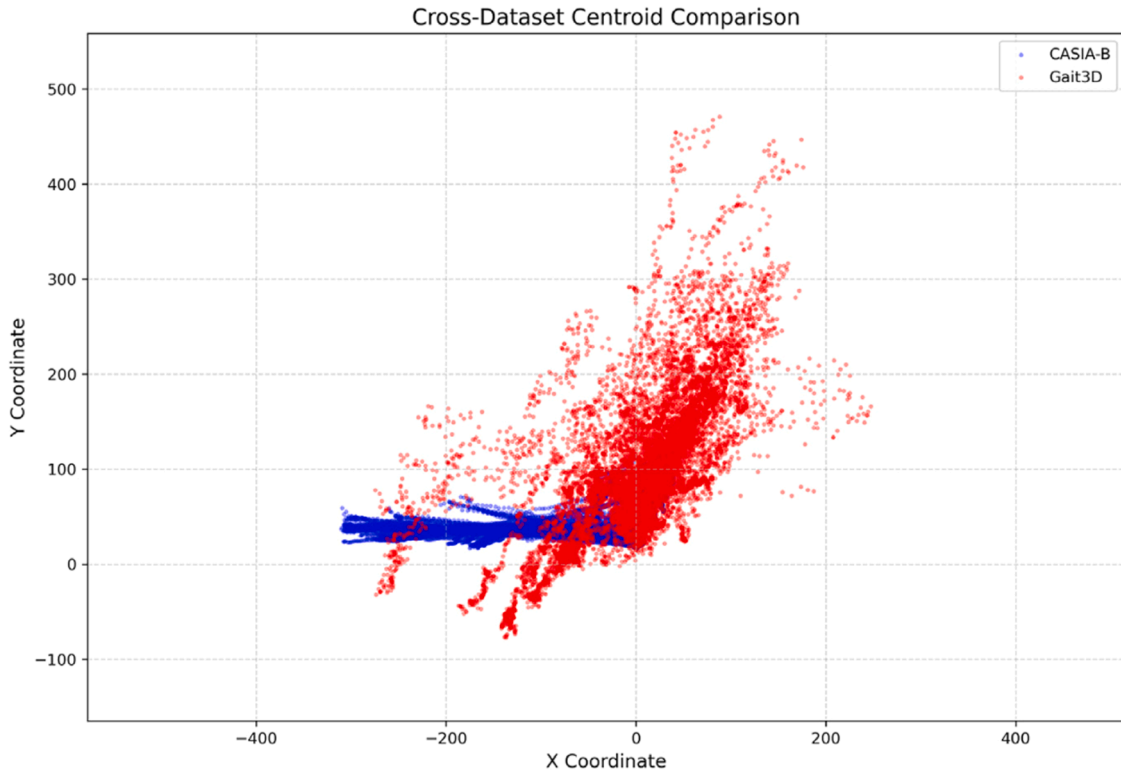


Fig. 9. Comparison of the center points of gait feature distributions across datasets.

model across different datasets, this paper conducts a cross-dataset visual comparison of gait features extracted from the CASIA-B and Gait3D datasets. In the figure, blue points represent the distribution of gait features from the CASIA-B dataset, while red points represent the distribution of features from the Gait3D dataset. It can be observed that there are significant differences in the distributions of the two datasets in the feature space. The feature distribution of the CASIA-B dataset is relatively concentrated and exhibits a more stable structure, whereas the feature distribution of the Gait3D dataset is broader in range and more dispersed. This is primarily due to differences between the two datasets in terms of acquisition environments, camera layouts, variations in viewing angles, and walking conditions, which result in distinct distribution characteristics of gait features in high-dimensional space. In contrast, the CASIA-B dataset is typically collected in relatively controlled experimental environments, resulting in a more compact feature distribution. The Gait3D dataset, however, incorporates more complex real-world factors, such as varying background environments, lighting conditions, and a wider range of pedestrian behavior patterns, leading to greater complexity and uncertainty in its feature distribution. From a modeling perspective, these cross-dataset distribution differences also reflect the cross-domain challenge commonly encountered in gait recognition tasks, where a model trained on one dataset may face feature distribution shifts when applied to another dataset.

As shown in Fig. 10, this paper analyzes the impact of different feature fusion operations and attention mechanisms on gait recognition performance under varying angle differences. The horizontal axis in the figure represents the angle difference, while the vertical axes represent the Rank-1 accuracy and mAP values, respectively. The overall trend shows that as the angle difference gradually increases, the recognition performance of all methods declines to some extent, indicating that angle variation remains a significant challenge in the gait recognition task. However, there are significant differences in performance among

various operation strategies. For simple element-wise operations (such as Element-wise Addition and Element-wise Multiplication), although they maintain high recognition rates under small angle differences, their performance declines more noticeably as the angle difference increases. In contrast, feature fusion methods based on average pooling demonstrate better stability under larger angle differences. Furthermore, in comparative experiments of attention mechanisms, it was observed that Multi-Head Attention (MHA) and the improved attention module achieved high Rank-1 and mAP values under various view angle conditions, with the improved attention fusion method maintaining the best performance across most ranges of view angle differences. This indicates that attention mechanisms can effectively capture key features in critical motion regions of the human body, thereby enhancing the model's discriminative capability under complex view angle variations. Further evaluate the robustness of the proposed model, misclassification cases are analyzed under challenging conditions such as viewpoint variations and clothing changes. It is observed that errors mainly occur under extreme cross-view scenarios or significant appearance variations. However, the block-based feature fusion strategy reduces these errors by combining complementary local and global features, thereby improving overall discrimination performance.

#### 4. Conclusion

This paper addresses the issue of degraded recognition performance in gait recognition tasks under complex environments, caused by factors such as changes in viewing angle, clothing, and carried items. We propose a gait recognition method that combines an attention-based convolutional network with block-based feature modeling. By integrating human appearance and motion information and utilizing deep learning models to effectively model gait features, this method enhances the robustness and recognition performance of gait recognition systems

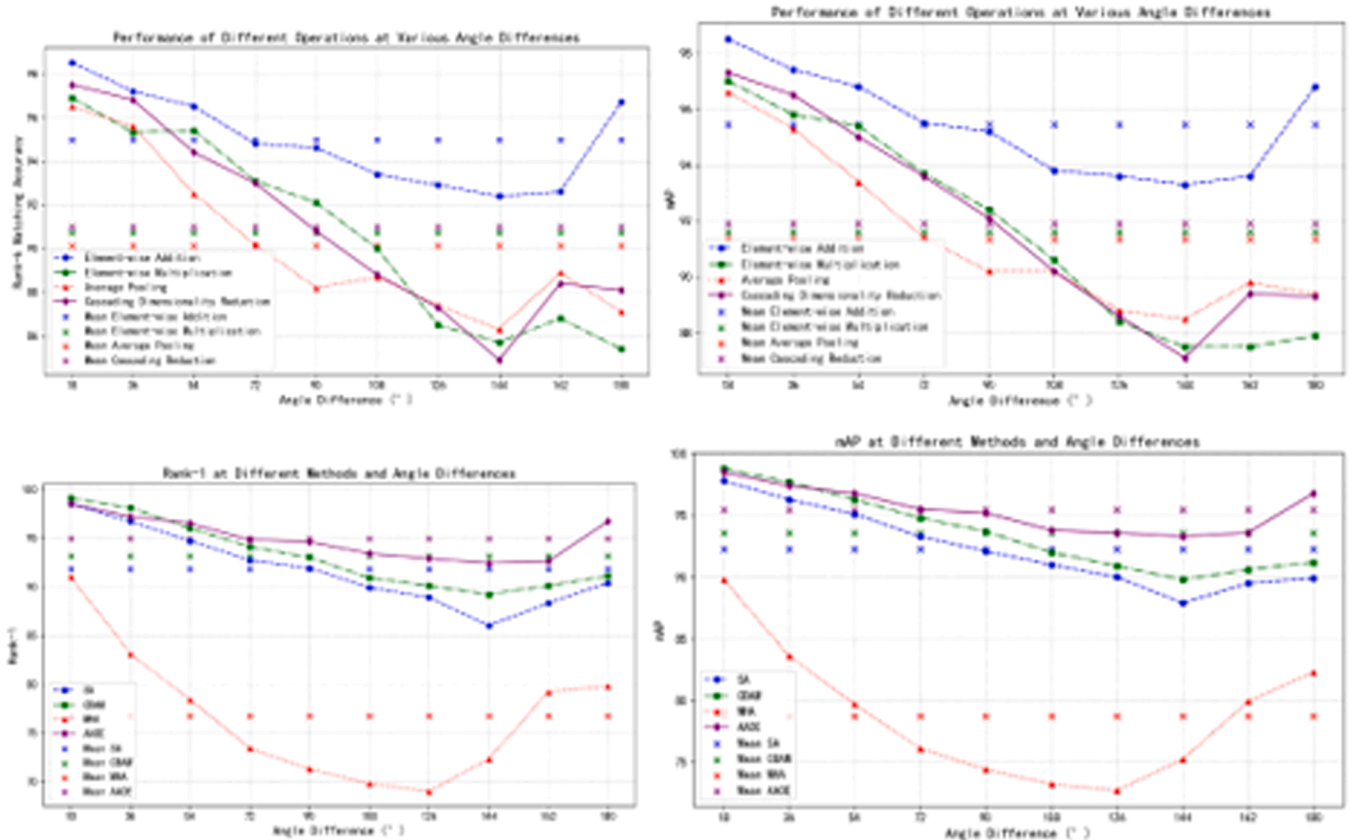


Fig. 10. Performance comparison of different feature fusion and attention mechanisms under varying view differences.

under complex conditions. Specifically, this paper first constructs a gait representation using gait contour sequences and combines it with optical flow information to extract the human motion region, thereby obtaining richer dynamic gait information. During the feature extraction stage, a dual-branch convolutional neural network architecture based on transfer learning is constructed, enabling the model to learn human appearance features and motion features separately, thus fully leveraging the complementary relationship between the two types of information. In the experimental section, the proposed method is systematically evaluated on the CASIA-B dataset, with validation conducted under various walking conditions (normal walking, carrying objects, and wearing a jacket) and different camera viewpoints. The experimental results demonstrate that the proposed method outperforms multiple baseline models on metrics such as Rank-1 accuracy and mAP. The developed architecture shows excellent results on the chosen dataset; nevertheless, the generalization of the developed algorithm to new data sets is an essential research topic for future work. While the incorporation of appearance and motion information improves the recognition rate, variations in the properties of the data set, like the distribution of viewpoints, environment, and variation in the subjects, could affect the performance of the method. Moreover, while the fusion of several networks offers higher precision, it entails greater computation requirements, making the process less efficient. Future work concentrates on developing a more efficient model and testing the proposed technique on multiple data sets.

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## Data availability

Not applicable.

## Ethical Approval

This article does not contain any studies involving human participants or animals performed by any of the authors.

## Consent to participate

Not applicable.

## Consent to publication

All authors have provided consent for publication of this manuscript.

## CRedit authorship contribution statement

**Lu Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation.

## Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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